

A Project Report
on
**Enhancing Medical Imaging Interpretation With Deep
Learning Models**

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MCT
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Certificate

This is to certify that

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Have satisfactory completed this project entitled

Enhancing Medical Imaging Interpretation With Deep Learning Models

Towards the partial fulfillment of the
BACHELOR OF ENGINEERING
IN
(COMPUTER ENGINEERING)
as laid by University of Mumbai.

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Project Report Approval for B. E.

This project report entitled "**Enhancing Medical Imaging Interpretation With Deep Learning Models**" by **Siddhi Avhad, Sanika Dhadve, Sarang Gajare, Purvaja Gangurde** is approved for the degree of Computer Engineering.

Examiners

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Date:

Place:

Declaration

We wish to state that the work embodied in this project titled "Enhancing Medical Imaging Interpretation With Deep Learning Models" forms our own contribution to the work carried out under the guidance of "Prof. Dilip Dalgade" at the Rajiv Gandhi Institute of Technology.

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission.

We understand that any violation of the above will be cause for disciplinary action by the institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

Medical imaging plays a crucial role in the early detection and diagnosis of life-threatening diseases, enabling timely treatment and improved patient outcomes. However, traditional diagnostic methods heavily rely on manual examination, which can be time-consuming and prone to human error. The emergence of deep learning (DL) models, particularly convolutional neural networks (CNNs), has significantly enhanced the field of medical imaging by offering more accurate, automated, and efficient disease classification. This study presents an advanced deep learning-based predictive model aimed at diagnosing three critical diseases—breast cancer, brain tumors, and kidney stones—using state-of-the-art CNN architectures. Our system is deployed as a real-time web-based platform, allowing seamless disease detection through medical image analysis. A comparative study of existing methodologies reveals various challenges in traditional and machine learning-based diagnostic approaches. Previous studies on breast cancer detection have used machine learning models such as Support Vector machines (SVM), Decision Trees, and basic CNN frameworks. While some models achieved high accuracy, they often lacked real-world validation on histopathological images, reducing their clinical applicability. Similarly, research on brain tumor detection using CNNs and transfer learning models like VGG16 and ResNet demonstrated promising results; however, many faced computational inefficiencies, high training times, and insufficient dataset diversity, impacting their generalization. For kidney stone classification, conventional machine learning models often struggled with ultrasound image noise, leading to lower accuracy and high false positive rates. While deep learning approaches have improved classification performance, previous studies lacked proper hyperparameter tuning and optimization techniques, making them less effective for practical deployment. Our proposed system addresses these gaps by implementing multiple CNN architectures, including CNN (Keras), VGGNet, MobileNetV2, and ResNet50, ensuring robust feature extraction and high classification accuracy. The model is trained using advanced optimization techniques, such as the Adam optimizer, varying learning rates, early stopping, and dropout layers, to prevent overfitting and improve generalization. Each disease classification model is designed for high specificity and sensitivity: breast cancer detection is performed using histopathological images to differentiate between benign and malignant tumors with high diagnostic accuracy; brain tumor classification is carried out by categorizing tumors into glioma, meningioma, pituitary tumors, and no tumor, ensuring precise differentiation between tumor types; kidney stone identification is conducted using ultrasound images to detect kidney anomalies, including cysts, tumors, stones, and normal kidney conditions. A key advantage of our study is the real-time deployment of the model as a web-based diagnostic tool, making it accessible to both medical professionals and patients. Unlike conventional diagnostic systems that require high-end computing resources, our cloud-integrated solution enables fast and accurate image classification without the need for expensive hardware. To validate model performance, we conducted extensive evaluations using metrics such as precision, recall, F1-score, and Matthews Correlation Coefficient (MCC). The results demonstrate exceptional accuracy across all three diseases, with the brain tumor model achieving up to 97% accuracy, kidney stone detection surpassing 97%, and breast cancer classification maintaining high sensitivity and specificity.

The findings of this study highlight the potential of deep learning in revolutionizing disease detection and medical diagnostics. Our model not only outperforms existing methods in accuracy and efficiency but also offers a practical, scalable, and clinically viable solution for automated disease classification. Future work will focus on expanding dataset diversity, improving model interpretability with explainable AI techniques, and integrating the system into existing healthcare infrastructures. By demonstrating high reliability, automation, and real-world applicability, this study sets a new benchmark for deep learning-driven medical imaging interpretation, paving the way for enhanced healthcare solutions worldwide.

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Chapter 1

Introduction

1.1 Introduction Description

Medical imaging is a key to the early diagnosis and detection of lethal diseases. Conventional diagnostic processes are based on manual inspection of medical scans, which can take time and is susceptible to human error. Incorporation of deep learning (DL) into healthcare has brought about new avenues of automating and enhancing disease detection, increasing diagnostic accuracy and efficacy. In the research, we present a web-based diagnostic platform that employs deep learning models to detect breast cancer, brain tumors, and kidney stones using medical imaging. Breast cancer is among the most common and lethal conditions in women worldwide, enabling early diagnosis with histopathological imaging greatly improves survival. Histopathology images present microscopic images of tissue specimens, which facilitate proper identification of cancerous cells[5]. Our breast cancer detection model is trained on such images and classifies them into two categories: benign (non-cancerous) and malignant (cancerous). Similarly, brain tumors require early diagnosis for effective treatment. Using Magnetic Resonance Imaging (MRI) scans, the model categorizes brain tumors into four distinct classes—glioma, no tumor, meningioma, and pituitary—with studies demonstrating high accuracies in this domain [3]. Additionally, kidney stone disease (nephrolithiasis) is a common urological condition that, if untreated, can lead to severe complications. Diagnosis is traditionally performed using X-rays, ultrasound, MRI, and Computed Tomography (CT) scans, all of which require manual interpretation by radiologists. Our deep learning model automates kidney stone detection and classifies medical images into four categories: cyst, tumor, stone, and normal [1].

To develop the system, we curated and preprocessed three distinct medical image datasets, ensuring robust feature extraction. We employed Convolutional Neural Networks (CNN) with multiple layers to learn deep representations from these images and achieve high classification accuracy, with notable accuracies such as 84%, 97% and 97% for breast cancer, brain tumor and kidney stone detection respectively [4] demonstrating their effectiveness in real-world applications. Furthermore, confusion matrices were analyzed to evaluate model performance, ensuring reliability for clinical use. The prime directive of this research is to develop a scalable, effortless, and fully automated diagnostic system that leverages CNN to process medical images and generate real-time predictions. By integrating advanced deep learning architectures with web-based deployment, the platform ensures efficient feature extraction, high-throughput image analysis, and robust classification performance. Addition-

ally, the study examines key challenges and potential advancements in deep learning-driven diagnostic frameworks, focusing on dataset diversity, model robustness across varying imaging conditions, and seamless integration into clinical workflows.

Churn, which can also be termed customer attrition, is essentially a pivotal metric that unveils the number of customers who discontinue using a company's services or a particular product. By closely monitoring this metric, most firms endeavor to gain insight into the underlying causes of high churn rates and then formulate reactive action plans to address these issues. However, what if businesses had the ability to go beyond mere reactive responses? What if they could anticipate a specific customer's likelihood to sever ties with the company and, in doing so, pre-emptively take the necessary steps to prevent such a departure?

1.2 Organization of Report

- **Ch.1:** The role of deep learning in medical imaging interpretation and its significance in disease detection. It highlights the development of a deep learning-based web platform for diagnosing breast cancer, brain tumors, and kidney stones. The chapter also discusses the advantages of AI-driven automation in enhancing diagnostic accuracy and reducing human error.
- **Ch.2:** The section provides a comprehensive review of existing research on deep learning applications in medical imaging. It discusses various CNN architectures and machine learning techniques used for disease classification, along with key studies that have contributed to the advancement of AI-driven diagnostics for breast cancer, brain tumors, and kidney stones.
- **Ch.3:** The proposed deep learning-based diagnostic system. It outlines the methodology used for data collection, preprocessing, and model training. The integration of CNN architectures, feature extraction techniques, and web-based deployment is discussed to demonstrate the system's efficiency in real-time disease detection.
- **Ch.4:** The experimental setup, including dataset sources, preprocessing steps, and model training procedures. It describes the CNN architectures used, hyperparameter tuning strategies, and optimization techniques applied to enhance model performance. The training pipeline and computational requirements are also outlined.
- **Ch.5:** The performance evaluation of the deep learning models using metrics such as precision, recall, F1-score, and Matthews Correlation Coefficient. The classification results for breast cancer, brain tumors, and kidney stones are analyzed using confusion matrices and classification reports. The impact of model optimizations on diagnostic accuracy is also discussed.
- **Ch.6:** The final chapter summarizes the key contributions of this study in advancing medical imaging interpretation using deep learning. It highlights the effectiveness of CNN models in disease detection and discusses potential future improvements, including dataset expansion, model generalization, and clinical integration for real-world healthcare applications.

Chapter 2

Literature Review

A literature survey helps in understanding recent advancements in a field of study and plays a crucial role in shaping research by identifying gaps in existing studies. It provides an outline of performed activities by various authors and researchers. This section analyzes research related to the application of deep learning in medical imaging. Reviewing existing literature helps in understanding key concepts, exploring previous findings, and identifying areas for improvement. It also connects new research with existing data, ensuring a strong foundation for further advancements.

- [1] reviewed different ML and DL methods for kidney stone detection, such as Support Vector Machines (SVM), Random Forest, Decision Trees, and Convolutional Neural Networks (CNN). However, the research lacked experimental verification and did not introduce a new model.
- [2] proposed CNN-6, ResNet50, and VGG16-based models for kidney tumor detection, achieving accuracies of 97%, 96%, and 92%, respectively. The study emphasized the need for larger datasets to develop more versatile and practically useful models.
- [3] described the implementation of CNN, VGG-16, and ensemble learning approaches for MRI-based classification, achieving 98.14% accuracy. Despite its high performance, the study highlighted computational constraints and extended training times.
- [4] applied CNN for tumor detection in MRI scans, achieving 96% accuracy. However, the model was limited to binary classification (Tumor/No Tumor) and did not differentiate between tumor subtypes.
- [5] utilized a Deep Convolutional Neural Network (DCNN), achieving the highest accuracy of 89.23%. The study noted performance degradation due to class imbalance between benign and malignant cases.
- [6] compared ANN, CNN, SVM, Random Forest, KNN, and Logistic Regression for breast cancer forecasting. ANN demonstrated the highest accuracy of 99.3%, followed by CNN (97.3%) and SVM with Random Forest (96.5%). However, the study was based on structured datasets rather than real-world histopathology images, limiting its clinical applicability.
- [7] analyzed multiple deep learning models, including CNN, SVM, KNN, Naïve Bayes, and Decision Trees, for breast cancer detection. While the study provided valuable insights, it lacked real-world implementation and direct performance comparisons.

- [8] implemented CNN-based deep learning with transfer learning, achieving accuracies of 83.2% (surface), 90.4% (section), and 85.6% (mixed). The research faced challenges related to small datasets, class imbalance, and dependence on ImageNet for feature extraction.
- [9] investigated kidney stone detection using CNN, GLCM feature extraction, Fuzzy C-means, and Backpropagation Neural Networks (BPN), reporting an accuracy of 92.5%. However, the study encountered challenges such as noise in ultrasound images and a limited dataset.
- [10] conducted a study comparing deep learning models such as CNN, VGG16, VGG19, and InceptionV3 for brain tumor detection using medical images. VGG16 obtained the best accuracy of 98%, demonstrating its efficiency in tumor classification. The study noted that no data augmentation measures (e.g., rotation, cropping) were used, which may affect model generalization.

These studies collectively highlight the advancements in deep learning for medical diagnostics while also emphasizing existing challenges, including restricted datasets, class imbalance, high computational expenses, and the necessity of real-world testing. Future work should focus on improving dataset diversity and enhancing deep learning models for deployment in clinical practice.

2.1 Limitation Existing system or research gap

Despite significant advancements in deep learning for medical imaging, several challenges persist, limiting the real-world applicability of these models. In kidney stone detection, many studies lack experimental validation, rely on small datasets, and struggle with noise in ultrasound images, affecting accuracy. Similarly, the brain tumor detection model, while achieving high accuracy, faces computational constraints, extended training times, and limited classification scope, often being restricted to binary classification rather than identifying multiple tumor subtypes. Breast cancer detection research also encounters issues such as class imbalance, dependence on structured datasets rather than real-world histopathology images, and a lack of practical implementation in clinical settings.

These challenges highlight key research gaps, including the limited availability of diverse and balanced datasets, the computational complexity of deep learning models, and issues with overfitting, which reduce model generalization to new data. Additionally, many existing models focus on binary classification instead of multi-class identification, restricting their diagnostic capability. Furthermore, most studies lack real-world testing and clinical integration, making it difficult to deploy these models effectively in hospitals. Addressing these limitations by developing larger, more diverse datasets, optimizing models for efficiency, and validating performance in real-world medical workflows will be crucial for enhancing deep learning's impact in healthcare diagnostics.

2.2 Problem Statement and objectives

Problem Statement

Accurate interpretation of medical images is crucial for identifying ailments such as kidney stones, brain tumors, and breast cancer. Standard approaches depend on radiologists conducting manual analyses, which are inefficient and prone to errors. Existing deep learning models often rely on basic CNNs that are not robust, scalable, or adequately evaluated—particularly when applied to imbalanced datasets. This limits their effectiveness in real clinical environments. An advanced, automated diagnostic system with high accuracy across various imaging tasks is needed. This project addresses these challenges by using optimized deep learning architectures such as ResNet50, MobileNet, and custom CNNs. It incorporates balanced performance metrics and a web-based platform to improve accessibility and clinical adoption, enabling faster, more reliable, and scalable medical image interpretation.

Objectives:

- To design an automated deep learning-based system for accurate interpretation and diagnosis using medical imaging data.
- To analyze and benchmark the performance of different deep learning architectures for medical image classification using standard evaluation metrics.
- To develop a user-friendly web-based interface for real-time diagnostic support and potential integration into clinical workflows.

Chapter 3

Proposed System

The proposed system is an elaborate deep learning-based medical diagnostic platform meant for spotting and classifying diseases utilizing medical imaging modalities like MRI, CT scans, and histopathological images. It centers around three key diagnostic tasks: brain tumor detection, kidney stone classification, and breast cancer identification. The system starts with data gathering and preprocessing labeled datasets confirmed by clinical experts to ensure clinical reliability. Each dataset undergoes scaling, resizing, and augmentation techniques—such as rotation, shifting, and flipping—to enhance model generalization. After this, the datasets are split into training, validation, and test sets with class balance preserved throughout. Individual CNN models are trained for each disease class, aided by transfer learning from the ResNet50 and MobileNetV2 architectures. These models include best practices such as batch normalization, dropout, L2 regularization, and global average pooling to ensure maximum accuracy and reduced overfitting.

The model optimization process is facilitated by methods like early stopping, learning rate scheduling, and checkpointing to maintain the highest-performing model. The models are tested extensively employing a battery of performance metrics like accuracy, precision, recall, F1-score, AUC-ROC, and Matthews Correlation Coefficient (MCC) to guarantee robustness across heterogeneous clinical settings. The system is tailored for smooth integration into healthcare settings, where input processed images are passed into trained models to provide high-confidence diagnostic predictions. By integrating high-quality, expert-approved data with deep learning methods and sound evaluation procedures, the system proposed here is expected to help healthcare professionals make early disease diagnoses and accurate diagnoses, ultimately leading to better patient outcomes.

3.1 Analysis/Framework/Algorithm

3.1.1 Framework

Frameworks are tools that help developers build applications, websites, and digital systems. They provide a structured environment with predefined libraries, components, and best practices, enabling efficient development and deployment.

In this project, we utilize deep learning frameworks such as TensorFlow and Keras to develop a neural network model for hyperglycemia detection. These frameworks offer powerful tools for model building, training, and evaluation, ensuring efficient handling of ECG and PPG data. The following components are integral to our framework:

- **TensorFlow:** Provides a flexible platform for implementing and optimizing deep learning models.
- **Keras:** A high-level neural network API running on TensorFlow, used for building and training deep learning models with ease.
- **Scikit-learn:** Used for preprocessing data, splitting datasets, and computing performance metrics such as confusion matrices and ROC curves.
- **Matplotlib:** Helps visualize model training progress, performance metrics, and feature distributions.
- **NumPy and Pandas:** Used for efficient numerical computation and dataset management.

By leveraging these frameworks, our system ensures accurate and real-time prediction of hyperglycemia, making it suitable for deployment in wearable healthcare applications.

3.1.2 Implemented Algorithm/Models

A. Convolutional Neural Network (CNN)

A **Convolutional Neural Network (CNN)** is a class of deep neural networks specifically designed to analyze visual imagery. CNNs are widely used in image classification, object detection, and image segmentation due to their ability to automatically extract hierarchical spatial features from input data.

A typical CNN architecture includes the following key layers:

- **Convolutional Layers:** Apply learnable filters to detect local features such as edges, textures, and patterns.
- **ReLU Activation Function:** Introduces non-linearity, allowing the network to learn complex representations.
- **Pooling Layers (Max/Average):** Reduce the spatial dimensions of feature maps while retaining important information.

- **Fully Connected Layers:** Perform high-level reasoning and classification based on the extracted features.
- **Softmax Layer:** Converts the final output into class probabilities for multi-class classification tasks.

CNNs are highly effective for medical image classification due to their ability to capture pixel-level patterns and discriminative features.

B. Residual Network (ResNet)

ResNet (Residual Network) is a deep learning architecture developed to overcome the degradation problem that occurs when training very deep neural networks. As the number of layers increases, accuracy may saturate or even degrade. ResNet introduces *skip connections* (also called residual connections), which allow the network to learn residual mappings instead of direct transformations.

A typical residual block is represented as:

$$y = F(x, \{W_i\}) + x \quad (\text{Eq.9 Residual Block})$$

Where:

- x is the input to the block,
- $F(x, \{W_i\})$ is the residual function learned by the block,
- y is the output of the block.

Advantages of ResNet:

- Facilitates the training of very deep networks (e.g., ResNet-50, ResNet-101).
- Helps avoid vanishing gradient issues by providing shortcut paths for gradient flow.
- Achieves state-of-the-art accuracy in image classification tasks.

In medical image analysis, ResNet has proven effective due to its depth and capacity to capture rich hierarchical features essential for disease classification.

C. MobileNet

MobileNet is a lightweight and computationally efficient deep neural network model designed for mobile and embedded vision applications. It replaces standard convolutions with *depthwise separable convolutions*, significantly reducing the number of parameters and computational cost.

A standard convolution is factorized into two operations:

- **Depthwise Convolution:** Applies a single filter per input channel.
- **Pointwise Convolution (1×1):** Combines the outputs of the depthwise layer.

This reduces the number of computations from:

$$\text{Standard: } D_K \times D_K \times M \times N \quad \text{to} \quad \text{MobileNet: } D_K \times D_K \times M + M \times N$$

(Eq_10 MobileNet Computation)

Where:

- D_K is the kernel size,
- M is the number of input channels,
- N is the number of output channels.

Advantages of MobileNet:

- Highly efficient and fast, ideal for deployment in low-resource environments.
- Comparable accuracy to larger models with significantly fewer parameters.
- Suitable for real-time applications like portable medical diagnostic tools.

In this project, MobileNet is employed for its ability to deliver accurate classification results with minimal computational overhead, making it suitable for practical use in medical imaging scenarios with limited resources.

3.2 Design Details

3.2.1 Detailed Design

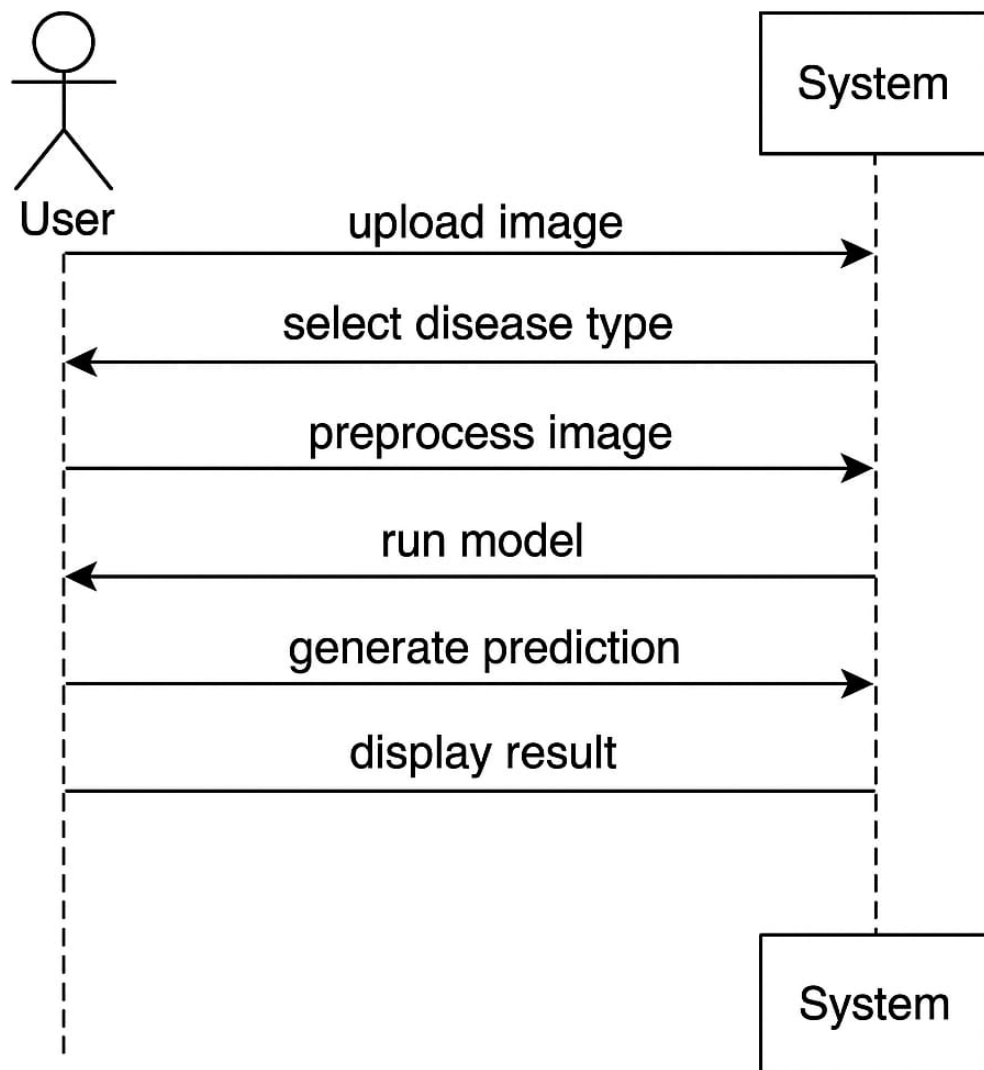


Figure 3.1: Use Case Diagram showing user interaction with the system

The diagram represents a sequential interaction where:

- The user begins by uploading an image.
- The system then prompts the user to select the disease type for analysis.
- The image undergoes preprocessing to make it suitable for model input.
- The model runs and generates a prediction based on trained data.
- Finally, the result is displayed to the user in an understandable format.

3.3 Methodology

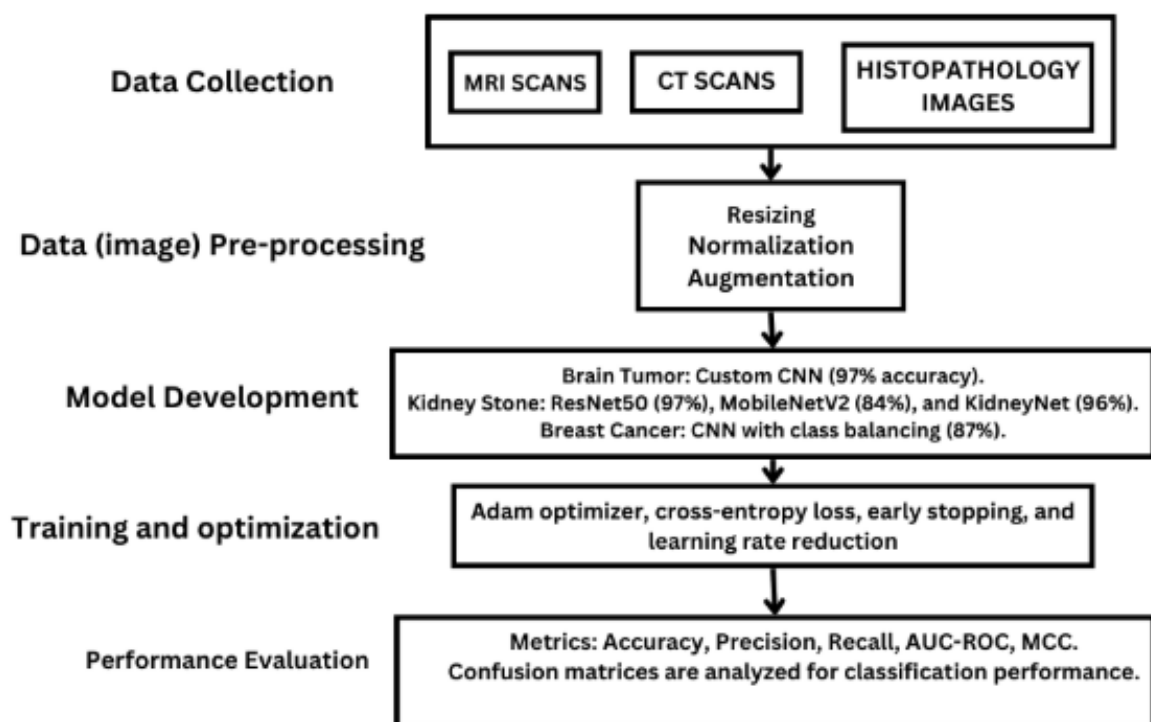


Figure 3.2: Proposed Methodology

A. Data Collection

Developing accurate deep learning (DL) models for medical diagnostics requires high-quality datasets. Medical imaging datasets, including CT scans, MRIs, and X-rays, are widely used for detecting diseases such as brain tumors, kidney stones, and breast cancer. However, a major challenge in this domain is the scarcity of labeled data, particularly for intricate scenarios. In response to this problem, dataset augmentation methods like flipping and scaling are used to improve the performance of models and enhance generalization.

A dataset of 7,023 MRI images was used for brain tumor detection, classified into glioma, meningioma, pituitary tumors, and no tumor. These MRI images were carefully labeled and verified by medical professionals to ensure accuracy. Similarly, in breast cancer detection, a dataset containing 277,524 50x50 image patches was created to identify Invasive Ductal Carcinoma (IDC). The dataset was labeled as malignant or benign, aiding in the automatic classification and grading of IDC regions to assess aggressiveness. For kidney stone detection, a dataset of 12,446 CT scan images was collected from hospitals in Dhaka, Bangladesh. These images were carefully labeled into four categories: cyst, tumor, stone, and normal, following expert verification by radiologists. The inclusion of verified medical datasets, combined with augmentation techniques, plays a crucial role in improving model accuracy, enabling early detection, and supporting clinical decision-making for better patient care.

B. Data Splitting and Sample Representation

The Kidney Stone, Breast Cancer, and Brain Tumor datasets were split into 80% training, 10% validation, and 10% testing for a balanced evaluation.

Kidney Stone Dataset:

Images were categorized into Cyst, Tumor, Stone, and Normal. The split ensured equal class representation, with 2,967 cyst, 1,826 tumor, 1,101 stone, and 4,061 normal images for training. The remaining were divided for validation and testing.

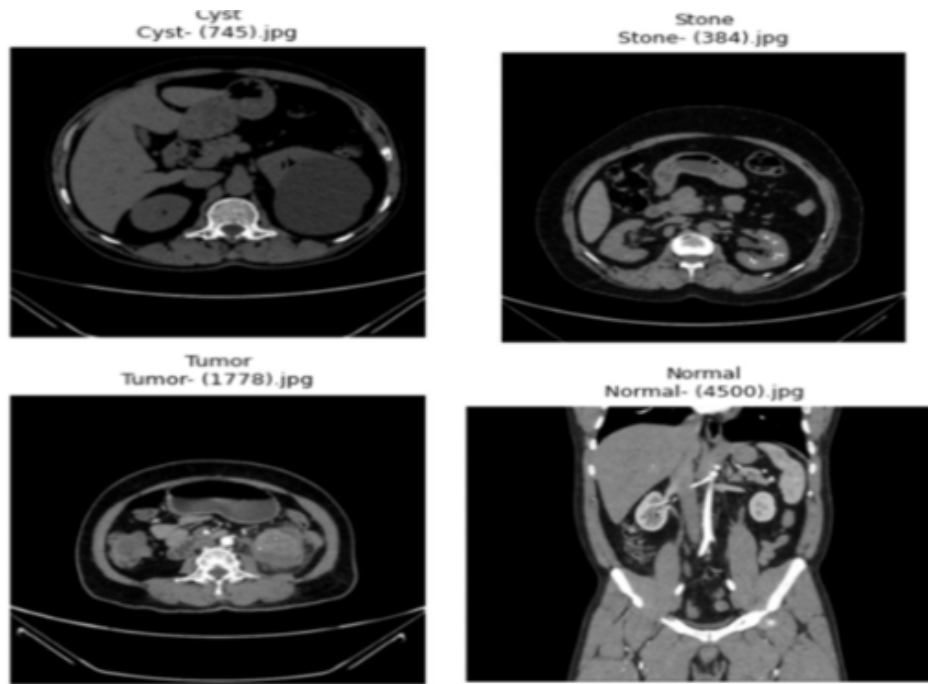


Figure 3.3: Sample Images - Kidney Dataset (Top Left: Cyst, Top Right: Stone, Bottom Left: Tumor, Bottom Right: Normal)

Brain Tumor Dataset:

Initially split 80% for training and 20% for testing, with further refinement into 90% training and 10% validation. The training set included 1,457 pituitary, 1,595 no-tumor, 1,339 meningioma, and 1,321 glioma images. Testing had 300 pituitary, 405 no-tumor, 306 meningioma, and 300 glioma images.

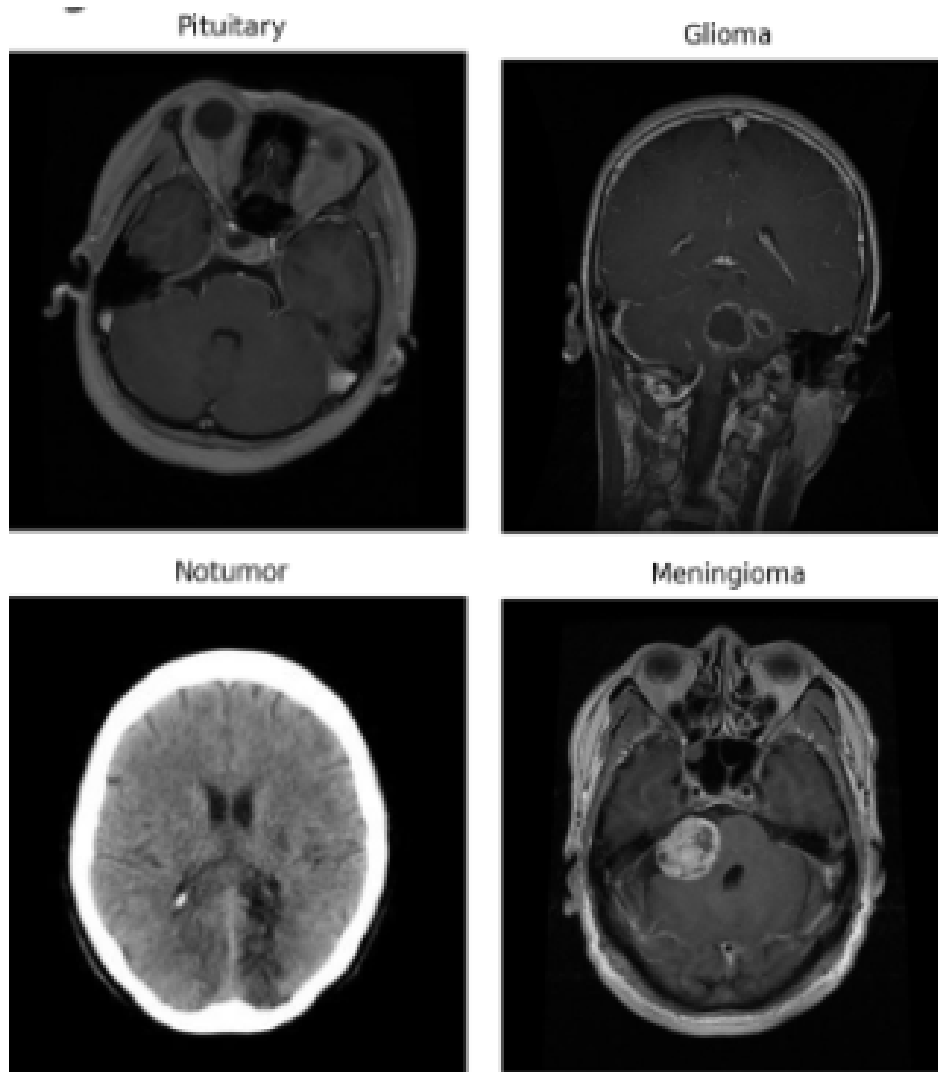


Figure 3.4: Sample Images - Brain Tumor Dataset (Top Left: Pituitary, Top Right: Glioma, Bottom Left: No Tumor, Bottom Right: Meningioma)

Breast Cancer Dataset:

The training set included 72,633 malignant and 183,361 benign images; the validation set had 12,195 malignant and 30,525 benign images; and the testing set contained 28,518 malignant and 71,563 benign images.

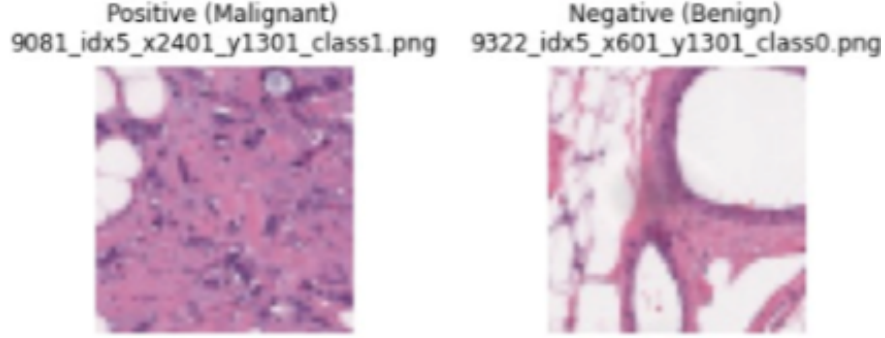


Figure 3.5: Sample Images - Breast Cancer Dataset (Left: Malignant, Right: Benign)

C. Data Processing and Augmentation

Effective data preprocessing enhances model performance and generalization in deep learning. For the kidney stone dataset, images were rescaled to $[0, 1]$, resized to 64×64 pixels, and divided into 80% training, 10% validation, and 10% testing. Data augmentation for training included shearing (0.2), zooming (0.2), and horizontal flipping, while validation and testing sets were only rescaled. Images were categorized into subdirectories for loading. For the Breast Cancer Dataset, similar preprocessing was applied. Images were rescaled, resized to 48×48 pixels, and split (80% training, 10% validation, 10% testing). Augmentation included rotation (20°), zooming (5%), shifting (10%), shearing (0.05), and flipping. In the Brain Tumor Dataset, images were rescaled, resized to 150×150 pixels, and split identically. Augmentation included rotation (20°), zooming (5%), shifting (10%), shearing (0.05), and flipping, with validation/testing sets only rescaled. Images were stored in labeled subdirectories for efficient loading.

The data augmentation parameters are set up to enhance diversity in the training data. The images are scaled by a factor of $1/255$ during pixel value normalization between 0 and 1. Also, random rotations up to 20 degrees are added to mimic varied orientations. The images are also randomly zoomed with a range of 0.05, and horizontal and vertical shifts of up to 10% of the image size are added. A shear transformation with a range of 0.05 also distorts the images to simulate real-world variations. Horizontal and vertical flip are both enabled to add to the dataset's diversity, and the 'nearest' fill mode is employed to complete any gaps resulting from these transformations. These preprocessing and augmentation steps ensure robust medical image classification by improving model accuracy and generalization.

D. Model Training

Developed multiple deep learning models aimed at medical image classification, specifically for detecting brain tumors, kidney stones, and breast cancer. Each of these models has been carefully designed using convolutional neural networks (CNNs), incorporating advanced techniques such as batch normalization, dropout, and regularization to enhance accuracy while

minimizing the risk of overfitting. Approach involves using both custom-built CNN architectures and transfer learning with pre-trained models, ensuring that the models generalize well on unseen data. Additionally, implemented strategies such as class weighting to handle dataset imbalance, data augmentation to improve robustness, and various callback functions to optimize training performance.

In brain tumor classification, there are two models designed. The initial model is a CNN with a series of convolutional blocks having an accuracy of 97%, each preceded by batch normalization to regularize learning and max pooling to diminish spatial dimensions without sacrificing significant features. The filters in the convolutional layers increase successively from 32 to 512 so that the model can detect both low-level and high-level features in the brain MRI scans. Instead of using traditional flattening, employ a GlobalAveragePooling2D layer, which reduces overfitting by averaging feature maps instead of merely reshaping them.

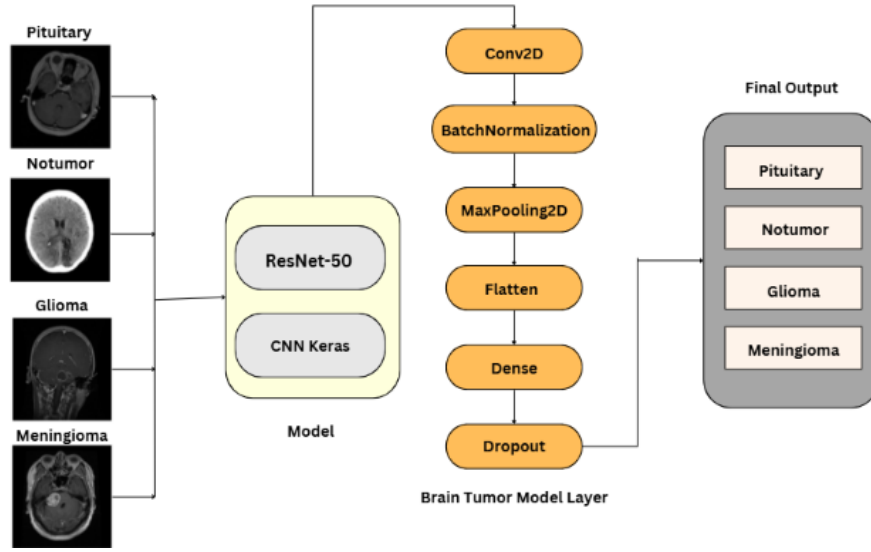


Figure 3.6: Brain Tumor Model

The fully connected layers that follow contain dropout to further enhance generalization, and the final softmax layer provides classification into one of the four brain tumor categories. Additionally, multiple callbacks, including EarlyStopping, which stops training if validation loss does not improve anymore; ModelCheckpoint, which stores the best model during training; and ReduceLROnPlateau, which reduces the learning rate if the model does not improve anymore, resulting in better convergence. The second brain tumor classification model follows a similar architecture but is slightly shallower, with fewer convolutional filters and layers, using a Flatten() layer instead of GlobalAveragePooling2D shown in below table.

The kidney stone classification combines three approaches: a transfer learning-based model using ResNet50 with accuracy 97%, MobileNetV2 with accuracy 84% and a custom CNN architecture called KidneyNet with accuracy 96%. In the transfer learning approach, fine-tuned a pre-trained ResNet50 model, freezing the initial layers to retain learned features while allowing deeper layers to be trained on the kidney stone dataset. Fine-tuning starts from layer 100 onward, enabling the model to adapt its feature extraction capabilities to kidney stone images.

Additional fully connected layers with dropout on top of ResNet50, ensures that the model can effectively classify different types of kidney stones. The second model, Kid-

Layer (type)	Output Shape	Param #
conv2d_8 (Conv2D)	(None, 48, 48, 32)	896
batch_normalization (BatchNormalization)	(None, 48, 48, 32)	128
conv2d_9 (Conv2D)	(None, 48, 48, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 48, 48, 64)	256
max_pooling2d_3 (MaxPooling2D)	(None, 24, 24, 64)	0
dropout_5 (Dropout)	(None, 24, 24, 64)	0
conv2d_10 (Conv2D)	(None, 24, 24, 64)	36,928
batch_normalization_2 (BatchNormalization)	(None, 24, 24, 64)	256
conv2d_11 (Conv2D)	(None, 24, 24, 128)	73,856
batch_normalization_3 (BatchNormalization)	(None, 24, 24, 128)	512
max_pooling2d_4 (MaxPooling2D)	(None, 12, 12, 128)	0
dropout_6 (Dropout)	(None, 12, 12, 128)	0
conv2d_12 (Conv2D)	(None, 12, 12, 128)	147,584
batch_normalization_4 (BatchNormalization)	(None, 12, 12, 128)	512
conv2d_13 (Conv2D)	(None, 12, 12, 256)	295,168
batch_normalization_5 (BatchNormalization)	(None, 12, 12, 256)	1,024
max_pooling2d_5 (MaxPooling2D)	(None, 6, 6, 256)	0
dropout_7 (Dropout)	(None, 6, 6, 256)	0
conv2d_14 (Conv2D)	(None, 6, 6, 256)	590,080
batch_normalization_6 (BatchNormalization)	(None, 6, 6, 256)	1,024
conv2d_15 (Conv2D)	(None, 6, 6, 512)	1,180,160
batch_normalization_7 (BatchNormalization)	(None, 6, 6, 512)	2,048
dropout_8 (Dropout)	(None, 6, 6, 512)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 512)	0
dense_2 (Dense)	(None, 512)	262,656
dropout_9 (Dropout)	(None, 512)	0
dense_3 (Dense)	(None, 4)	2,052

Figure 3.7: Summary of Brain Tumor

neyNet, is a specialized CNN architecture with three convolutional blocks, each preceded by batch normalization and max pooling. In contrast to deeper models, KidneyNet is comparatively shallow but attains high accuracy through the use of L2 regularization, which prevents overfitting by discouraging large weight values in all layers. The final softmax layer ensures precise classification by outputting probabilities for the four kidney stone categories. Summarize of kidney stones is given in Table.

In the breast cancer classification, a CNN model using Keras is specifically designed for classifying histopathological images of breast cancer. The architecture uses three convolutional blocks with an increasing number of filters in order to extract useful features from the images. Batch normalization is used to stabilize learning, and dropout layers with values of 0.25 to 0.5 are appropriately placed in order to avoid overfitting. Unlike the previous models, this model uses GlobalAveragePooling2D to provide a more compact representation

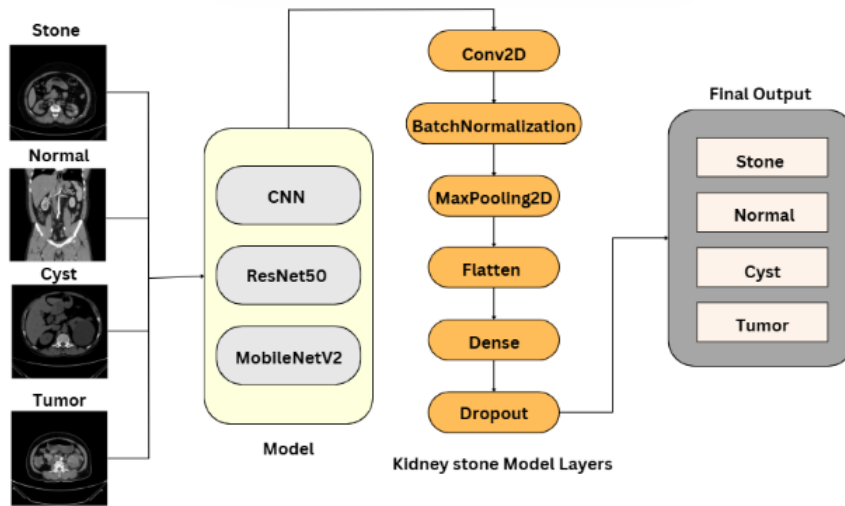


Figure 3.8: Kidney Stone Model

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_1[0][0]
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalization)	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalization)	(None, 56, 56, 64)	256	conv2_block1_1_conv[0]
conv2_block1_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_1_bn[0]
conv2_block1_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block1_1_relu[0]
conv2_block1_2_bn (BatchNormalization)	(None, 56, 56, 64)	256	conv2_block1_2_conv[0]
conv2_block1_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_2_bn[0]
conv2_block1_0_conv (Conv2D)	(None, 56, 56, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block1_2_relu[0]
conv2_block1_0_bn (BatchNormalization)	(None, 56, 56, 256)	1,024	conv2_block1_0_conv[0]
conv2_block1_3_bn (BatchNormalization)	(None, 56, 56, 256)	1,024	conv2_block1_3_conv[0]

Figure 3.9: Summary of Kidney Stone

of the learned features. The output layer has a softmax activation function, generating a final classification decision based on the learned image features.

In each of the models, hyperparameters such as the learning rate, batch size, and number of epochs are optimized to achieve the best results. Moreover, a grid search strategy is used to experiment with different combinations of hyperparameters, ensuring the models achieve

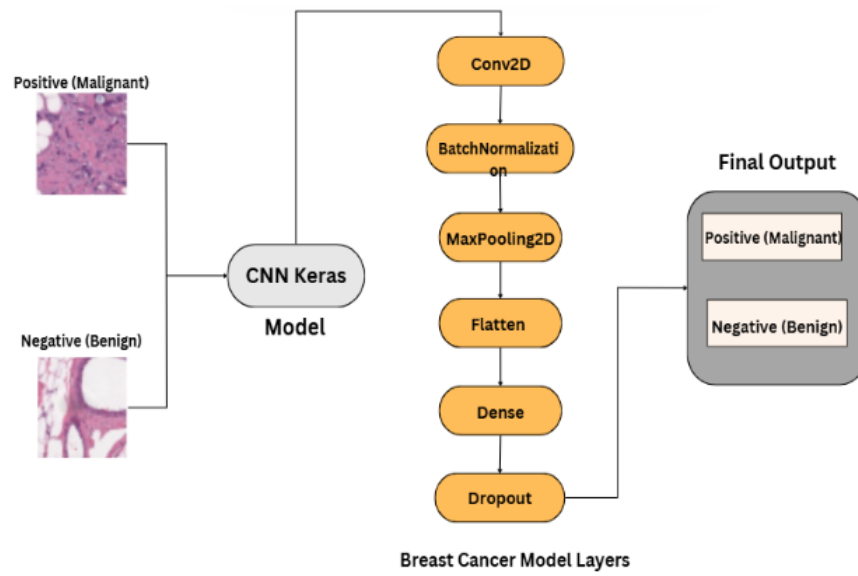


Figure 3.10: Breast Cancer Model

a robust performance on the given datasets.

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 46, 46, 32)	896
batch_normalization (BatchNormalization)	(None, 46, 46, 32)	184
max_pooling2d (MaxPooling2D)	(None, 23, 23, 32)	0
dropout (Dropout)	(None, 23, 23, 32)	0
conv2d_1 (Conv2D)	(None, 21, 21, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 21, 21, 64)	84
conv2d_2 (Conv2D)	(None, 19, 19, 64)	36,928
batch_normalization_2 (BatchNormalization)	(None, 19, 19, 64)	76
max_pooling2d_1 (MaxPooling2D)	(None, 9, 9, 64)	0
dropout_1 (Dropout)	(None, 9, 9, 64)	0
conv2d_3 (Conv2D)	(None, 7, 7, 64)	36,928
batch_normalization_3 (BatchNormalization)	(None, 7, 7, 64)	28
conv2d_4 (Conv2D)	(None, 5, 5, 64)	36,928
batch_normalization_4 (BatchNormalization)	(None, 5, 5, 64)	20
conv2d_5 (Conv2D)	(None, 3, 3, 64)	36,928
batch_normalization_5 (BatchNormalization)	(None, 3, 3, 64)	12
max_pooling2d_2 (MaxPooling2D)	(None, 1, 1, 64)	0
dropout_2 (Dropout)	(None, 1, 1, 64)	0
flatten (Flatten)	(None, 64)	0
dense (Dense)	(None, 256)	16,640
batch_normalization_6 (BatchNormalization)	(None, 256)	1,024
dropout_3 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 2)	514

Figure 3.11: Summary of Breast Cancer

Adoption of the best, optimal models is done to ensure robust training and evaluation. To counteract overfitting, dropout layers, L2 regularization, and global average pooling are applied, which minimize the number of trainable parameters and prevent the model from memorizing the training data. Learning rate reduction strategies were implemented to fine-tune weights for optimal performance. The models were trained by monitoring validation loss and accuracy, and testing was conducted on unseen data to ensure real-world applicability. Advanced techniques like early stopping and model checkpoints automatically save the best-performing model during training, preventing overfitting and saving time.

The combination of custom CNN architectures and transfer learning proved to be highly effective for medical image classification tasks. By incorporating best practices like data augmentation, class balancing, dropout, and adaptive learning rate scheduling, the models performed optimally on complex medical datasets. These models have the potential to significantly aid in the early detection of brain tumors, kidney stones, and breast cancer, thus contributing to improved diagnostic outcomes in medical imaging.

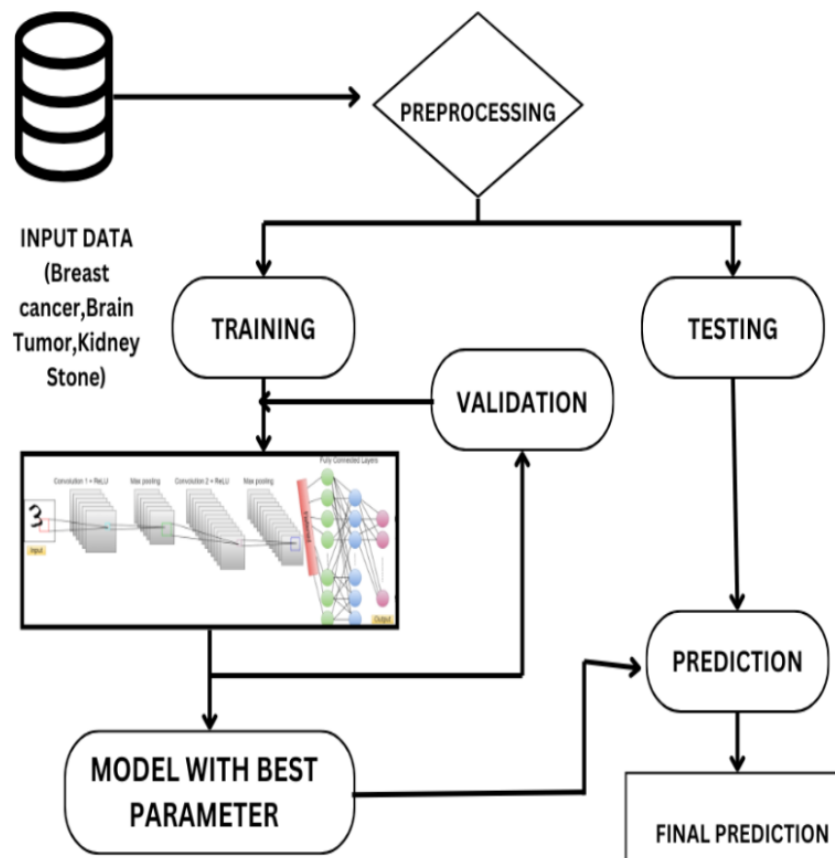


Figure 3.12: Flow of CNN Model

The flow of the CNN model begins with input image preprocessing, which includes resizing and normalization to a fixed size and value range. Then, the convolutional layers perform feature extraction, which is followed by pooling layers to reduce dimensionality. The features are passed through fully connected layers, followed by the softmax layer for classification. Dropout and batch normalization are applied to prevent overfitting and improve generalization.

E. Performance Evaluation

Several performance measures are used regularly to contrast the performance of a system. These include Accuracy, Precision, Recall, F1-Score, Area Under the Curve (AUC), Matthews Correlation Coefficient (MCC), and the Receiver Operating Characteristic (ROC) Curve. Below, each measure is explained, along with their mathematical representations.

Confusion Matrix

A confusion matrix is a two-dimensional table that summarizes the results of the classification performed and helps compare correct and incorrect predictions. It consists of four components:

- True Positives (TP): Correctly predicted positive cases.
- True Negatives (TN): Correctly predicted negative cases.
- False Positives (FP): Negative cases incorrectly classified as positive.
- False Negatives (FN): Positive cases incorrectly classified as negative.

Accuracy

Accuracy measures the overall percentage of correct predictions. It is calculated as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.1)$$

Accuracy is a useful measure when the classes are well-balanced, but can be misleading in the case of imbalanced datasets.

Precision (Positive Predictive Value)

Precision measures the proportion of positive predictions that are actually correct. It is calculated as:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.2)$$

Precision is important in scenarios where false positives are costly, such as in medical diagnoses.

Recall (Sensitivity, True Positive Rate)

Recall measures the proportion of actual positives that are correctly identified. It is calculated as:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.3)$$

Recall is particularly valuable when the loss of a positive case is costly, such as when diagnosing diseases.

F1-Score

The F1-score is the harmonic mean of Precision and Recall. It balances both Precision and Recall, and is particularly useful when dealing with imbalanced datasets. The formula is:

$$\text{F1-Score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.4)$$

Area Under the Curve (AUC)

The Area Under the Receiver Operating Characteristic (ROC) Curve (AUC) is a measure of the trade-off between sensitivity (Recall) and specificity. A higher AUC indicates a better model. AUC is calculated by plotting the True Positive Rate (TPR or Recall) against the False Positive Rate (FPR). The formula for AUC is:

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d\text{FPR} \quad (3.5)$$

Where:

$$\text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (3.6)$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}} \quad (3.7)$$

Matthews Correlation Coefficient (MCC)

The Matthews Correlation Coefficient (MCC) is a measure of the quality of binary classification, providing a score between -1 (bad) and +1 (ideal), with 0 indicating random predictions. It is calculated as:

$$\text{MCC} = \frac{\text{TP} \cdot \text{TN} - \text{FP} \cdot \text{FN}}{\sqrt{(\text{TP} + \text{FP})(\text{TP} + \text{FN})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}} \quad (3.8)$$

MCC is particularly useful for imbalanced datasets as it considers all four components of the confusion matrix.

Receiver Operating Characteristic (ROC) Curve

The ROC curve plots the True Positive Rate (TPR) against the False Positive Rate (FPR) at various threshold settings. The ROC curve helps determine the optimal decision threshold for classification. The ROC curve is a graphical representation of the model's ability to discriminate between positive and negative classes.

Chapter 4

Experimental Setup

4.1 Details of Database or Input to System or Selected Data

4.1.1 Breast Cancer Dataset

The breast cancer dataset for Invasive Ductal Carcinoma (IDC) detection consists of 277,524 image patches, each measuring 50×50 pixels in RGB format, refer These patches were extracted from whole mount histopathology slides of breast cancer tissue and manually labeled by expert pathologists. The dataset is divided into 198,738 IDC-negative patches (non-cancerous) and 78,786 IDC-positive patches (cancerous), ensuring a diverse representation of tissue samples. The images capture detailed cellular structures to aid in IDC classification and grading.

4.1.2 Brain Tumor Dataset

Brain tumors are one of the most serious neurological conditions, requiring accurate and early diagnosis for effective treatment. Proper classification of brain tumors helps in treatment planning and improves patient survival rates. The brain tumor dataset consists of 7,023 MRI images of the human brain, categorized into four classes: glioma, meningioma, no tumor, and pituitary tumor (refer Figure ??). These MRI scans serve as a crucial tool for detecting and diagnosing brain tumors, as Magnetic Resonance Imaging (MRI) is the most effective technique for identifying abnormalities in brain tissues. Each image is labeled accordingly to aid in accurate classification and segmentation tasks.

4.1.3 Kidney Stone Dataset

Kidney stones are solid deposits of minerals and salts that form in the kidneys and can cause severe pain and complications if untreated. They vary in size and composition, with common types including calcium oxalate, uric acid, struvite, and cystine stones. Early detection through imaging techniques like CT scans and ultrasounds is essential for effective treatment. The kidney stone dataset, collected from hospitals in Dhaka, Bangladesh, includes 1,377 CT scan images, aiding in automated detection and classification (refer FigureThis dataset helps in training AI models for improved diagnosis, treatment planning, and early intervention in kidney stone cases.

4.2 Preprocessing on Dataset

4.2.1 Breast Cancer Preprocessing

- **Rescaling:** Pixel values were rescaled by dividing by 255, ensuring that the values lie between 0 and 1 for better convergence during training.
- **Data Augmentation:**
 - **Training:**
 - * Rotation Range: Up to 20 degrees.
 - * Zoom Range: Within 5% of the image size.
 - * Width and Height Shift: Up to 10% shift.
 - * Shear Range: 0.05.
 - * Flipping: Horizontal and vertical.
 - **Validation and Testing:** Only rescaling was applied to preserve consistency and avoid introducing variations during evaluation.
- **Image Size Standardization:** All images resized to 48x48 pixels to match CNN input requirements.
- **Training, Validation, and Test Split:** 80% for training, 10% for validation, and 10% for testing.
- **Class Labeling and Augmentation:**
 - Labels were assigned as follows: 1 = Malignant (Positive), 0 = Benign (Negative), derived from directory names.
 - Class weights were computed to address class imbalance.

4.2.2 Brain Tumor Preprocessing

- **Rescaling:** Each pixel was divided by 255 to normalize the input.
- **Data Augmentation:**
 - **Training:**
 - * Rotation Range: Up to 20 degrees.
 - * Zoom Range: 5%.
 - * Width and Height Shift: Up to 10%.
 - * Shear Range: 0.05.
 - * Flipping: Horizontal and vertical.
 - **Validation and Testing:** Only rescaling was applied to maintain evaluation integrity.
- **Image Size Standardization:** Images resized to 150x150 pixels.
- **Training, Validation, and Test Split:** 80% training, 10% validation, 10% testing.
- **Directory Structure:** Organized by class labels into subdirectories (e.g., "Brain Tumor", "Normal") for use with `ImageDataGenerator`.

4.2.3 Kidney Stone Preprocessing

- **Rescaling:** Pixel values divided by 255 to normalize them in the $[0, 1]$ range.
- **Data Augmentation:**
 - **Training:**
 - * Shear Range: 0.2.
 - * Zoom Range: 0.2.
 - * Horizontal Flip: Enabled.
 - **Validation and Testing:** Only rescaling was applied.
- **Image Size Standardization:** All images resized to 64x64 pixels.
- **Training, Validation, and Test Split:** 80% training, 10% validation, 10% testing.
- **Directory Structure:** Images placed in class-specific folders (e.g., "Cyst", "Tumor", "Stone", "Normal") to be loaded via `flow_from_directory`.

A pie chart is employed to visualize the correlation between features, and those features exhibiting strong correlations are subsequently excluded from the analysis.

4.3 Performance Evaluation

4.3.1 Confusion Matrix

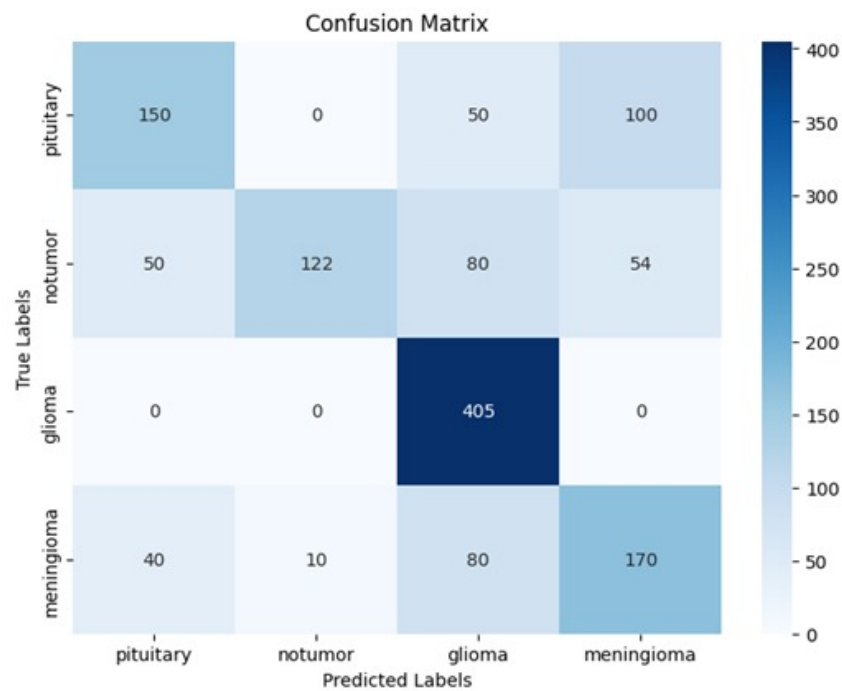


Figure 4.1: Confusion Matrix - Brain Tumor Model

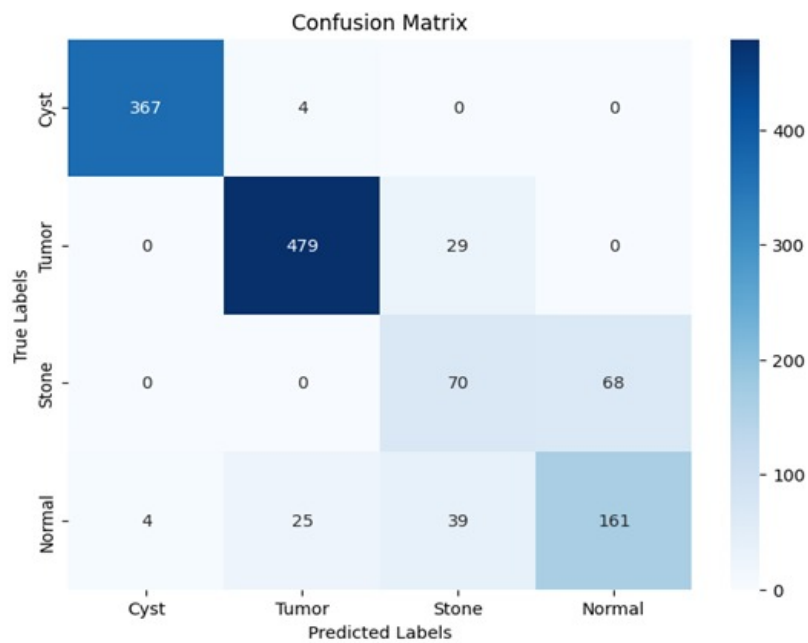


Figure 4.2: Confusion Matrix - Kidney Stone Model

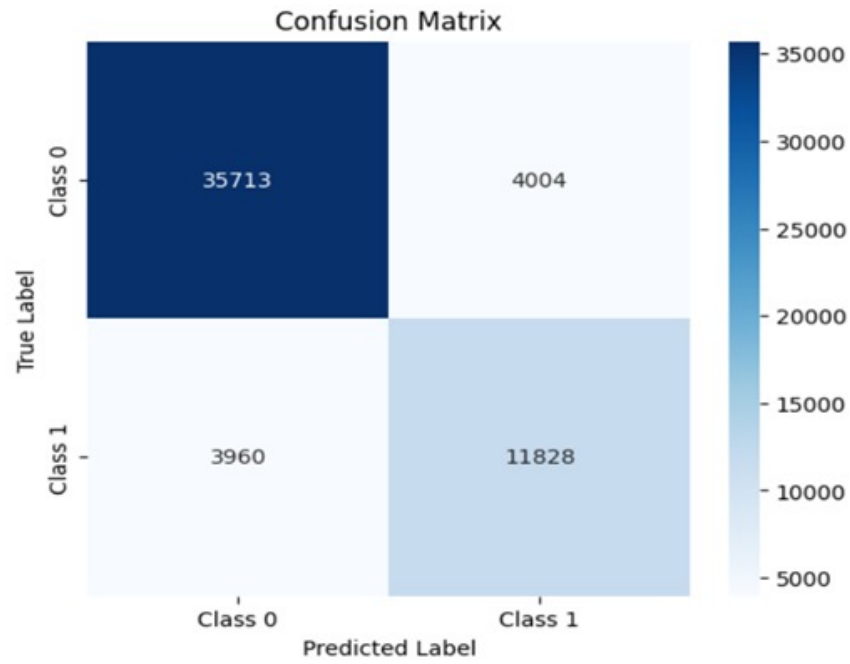


Figure 4.3: Confusion Matrix - Breast Cancer Model

4.3.2 AUC-ROC Curve

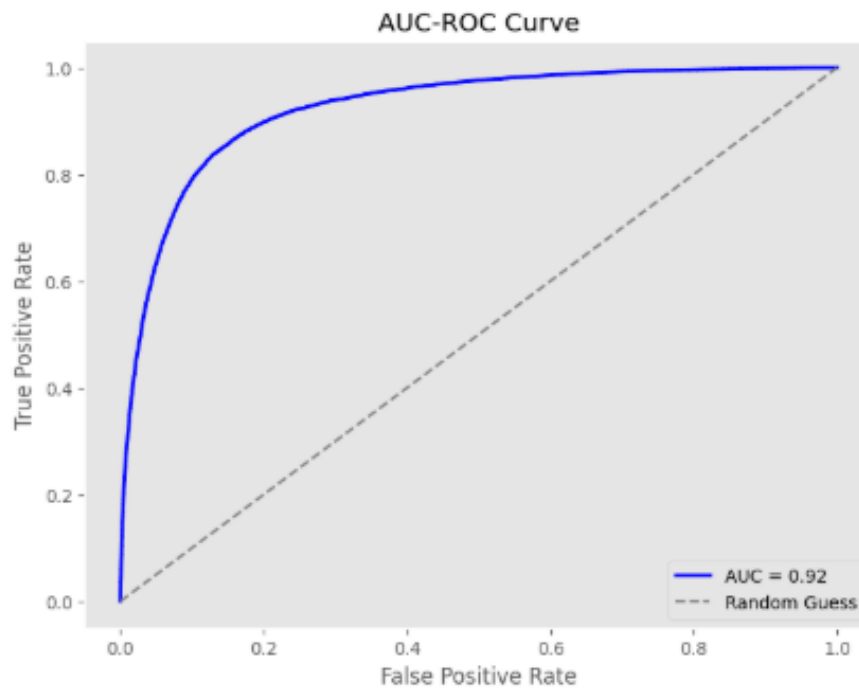


Figure 4.4: AUC-ROC Curve - Breast Cancer Model

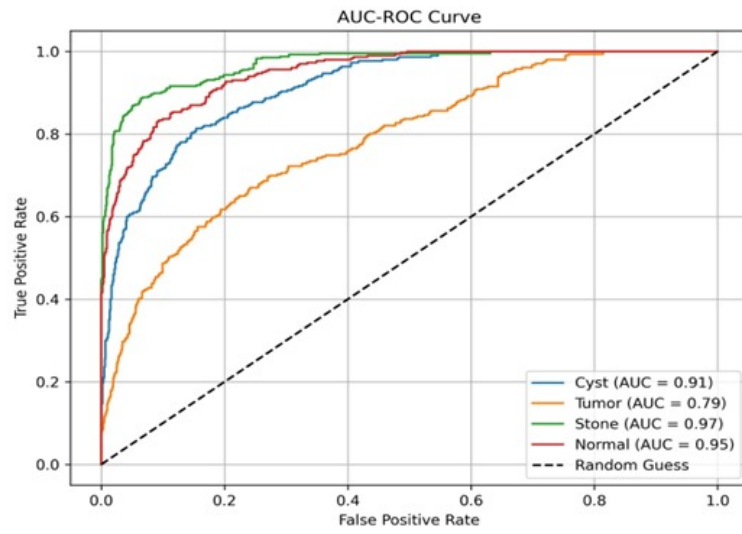


Figure 4.5: AUC-ROC Curve - Brain Tumor Model

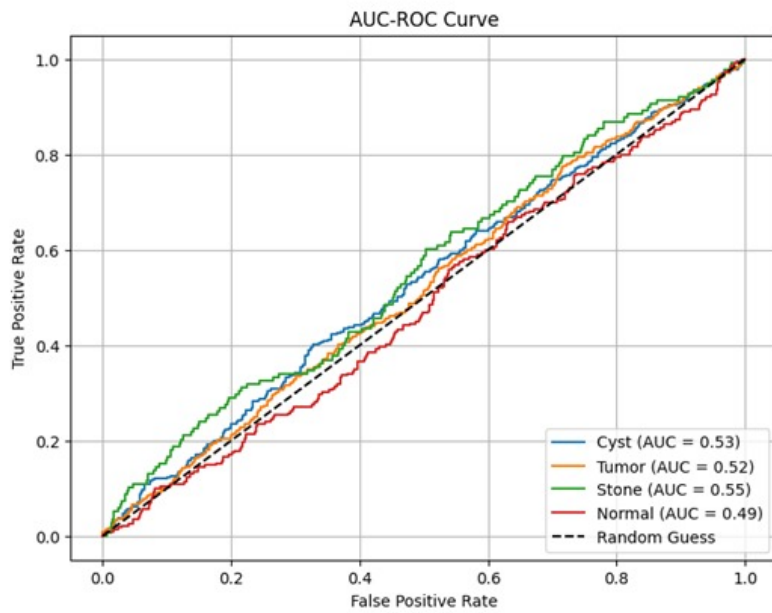


Figure 4.6: AUC-ROC Curve - Kidney Stone Model

4.4 Software and Hardware Set up

4.4.1 Software Setup

- **Programming Languages:** Python is the primary language used for deep learning model development and implementation.
- **Deep Learning and Machine Learning Libraries:**
 - TensorFlow: Open-source deep learning framework used for building, training, and deploying models.
 - Keras: High-level API for TensorFlow, simplifying neural network implementation.
 - PyTorch: Alternative deep learning library (used optionally for experimentation).
 - Scikit-Learn: Used for traditional machine learning tasks, feature extraction, and evaluation metrics.
- **Image Processing and Data Handling:**
 - OpenCV: For image preprocessing, resizing, and augmentation.
 - Pillow (PIL): For image manipulation and medical image handling.
 - NumPy: For handling large numerical datasets and image arrays.
 - Pandas: For structured data analysis, such as patient metadata.
- **Model Training and Evaluation Tools:**
 - Google Colab: Cloud-based Jupyter Notebook with GPU support for training deep learning models.
 - Jupyter Notebook: Interactive environment for developing and testing ML/DL models.
 - Matplotlib and Seaborn: For visualizing training metrics, confusion matrices, and performance graphs.
- **Web-Based Deployment and UI Frameworks:**
 - Streamlit: Python-based framework for building interactive diagnostic web applications.
 - Spyder (Streamlit): Used for front-end application development and deployment for disease detection.
- **Model Optimization and Deployment Tools:**
 - TensorFlow Lite: For optimizing models to run on mobile and edge devices.
- **Development Environment and Package Management:**
 - Anaconda: Provides a complete Python environment with libraries for deep learning, preprocessing, and training.

- **Version Control and Collaboration:**

- Git and GitHub: For version control, code management, and team collaboration.

- **Additional Tools Summary:**

- TensorFlow and Keras: Model building and training.
- OpenCV and PIL: Image processing and augmentation.
- Pandas and NumPy: Data handling and preprocessing.
- Matplotlib and Seaborn: Visualization of model performance.

4.4.2 Hardware Setup

- **Processor (CPU):** Intel Core i5 (11th Gen) or AMD Ryzen 7
- **RAM:** 16 GB DDR4
- **Storage:** 512 GB SSD (for faster data read/write operations)
- **Graphics Processing Unit (GPU):** NVIDIA GTX 1660 Ti / RTX 3060 (6GB+ VRAM)
- **Operating System:** Windows 10 / Ubuntu 20.04 LTS
- **Display:** 1080p Full HD Monitor

Chapter 5

Results & Discussions

5.1 Implemented Algorithm's Code

app.py (Main File)

BEGIN

```
IF "Breast Cancer" selected THEN
    SHOW info, IF button clicked THEN SWITCH to "BREAST.py"
```

```
ELSE IF "Brain Tumor" selected THEN
    SHOW info, IF button clicked THEN SWITCH to "BRAIN.py"
```

```
ELSE IF "Kidney Disease" selected THEN
    SHOW info, IF button clicked THEN SWITCH to "KIDNEY.py"
```

END

BRAIN.py

BEGIN

```
SET title "Brain Tumor Detection"
```

```
# Load model
```

```
FUNCTION load_model():
    RETURN trained model (Brain_tumor.h5)
```

```
# Preprocess image
```

```
FUNCTION preprocess_image(image):
    RESIZE to (224,224), NORMALIZE, ADD batch dimension
    RETURN processed image
```

```
# Validate MRI scan
```

```
FUNCTION is_grayscale(image):
    CHECK if RGB channels are nearly identical
    RETURN True if grayscale, else False
```

```

FUNCTION is_circular_or_oval(image):
    APPLY blur & edge detection
    CHECK for circular/oval shape using Hough Transform & contours
    RETURN True if valid shape detected

# Predict tumor type
FUNCTION predict_image(image):
    PROCESS image, GET prediction
    RETURN class label & confidence score

# Streamlit UI
DISPLAY file uploader (jpg/jpeg/png)

IF user uploads image THEN
    DISPLAY uploaded image
    IF not grayscale OR not circular/oval THEN
        SHOW error "Invalid MRI scan"
    ELSE
        IF "Predict" clicked THEN
            DISPLAY Prediction & Confidence
            SHOW prevention tips & symptoms
        ELSE
            PROMPT user to upload MRI scan
    END
END

BREAST.py

BEGIN
    SET application title "Breast Cancer Detection"

    # Load pre-trained model
    FUNCTION load_model():
        RETURN trained deep learning model (cancer_detection_model.h5)

    # Image preprocessing function
    FUNCTION preprocess_image(image):
        RESIZE image to (48,48)
        NORMALIZE pixel values between [0,1]
        ADD batch dimension
        RETURN processed image

    # Validate if the uploaded image is a histopathology scan
    FUNCTION is_valid_histopathology_image(image):
        CONVERT image to RGB array
        COMPUTE average color values

```

```

    IF average RGB values do not match expected histopathology colors THEN
        RETURN False
    CONVERT image to HSV color space
    APPLY color filter to detect pink/purple stain regions
    CALCULATE percentage of pixels matching stain color
    IF match percentage < 15% THEN RETURN False
    CONVERT image to grayscale and calculate texture complexity
    RETURN True if texture complexity > 15

# Function to predict disease from image
FUNCTION predict_image(image):
    PROCESS image using preprocess_image()
    GET prediction from trained model
    IDENTIFY class with highest probability (Benign/Malignant)
    RETURN class label and confidence percentage

# Streamlit UI Setup
DISPLAY file uploader for image (jpg/jpeg/png)

IF user uploads an image THEN
    DISPLAY uploaded image (resized for UI)
    IF image is not a valid histopathology scan THEN
        DISPLAY error message "Invalid Image!"
    ELSE
        IF user clicks "Predict" button THEN
            CALL predict_image()
            DISPLAY Prediction result and confidence score
            DISPLAY prevention measures
            DISPLAY common symptoms

ELSE
    DISPLAY "Please upload an image to proceed" message

END

KIDNEY.py

BEGIN
    SET title "Kidney Stone Detection"

    # Load model
    FUNCTION load_model():
        CHECK if model file exists, LOAD and RETURN model

    # Preprocess image
    FUNCTION preprocess_image(image):
        RESIZE to (224,224), NORMALIZE, ADD batch dimension, RETURN image

```

```

# Validate kidney scan
FUNCTION is_grayscale(image):
    CHECK if RGB channels are nearly identical, RETURN True/False

FUNCTION is_circular_or_oval(image):
    APPLY blur & edge detection, DETECT circles/ovals, RETURN True/False

# Predict kidney condition
FUNCTION predict_image(image):
    PROCESS image, GET prediction, RETURN class label & confidence

# Streamlit UI
DISPLAY file uploader (jpg/jpeg/png)

IF user uploads image THEN
    DISPLAY uploaded image
    IF not grayscale OR not circular/oval THEN
        SHOW error "Invalid Kidney Scan"
    ELSE
        IF "Predict" clicked THEN
            DISPLAY Prediction & Confidence
            SHOW prevention tips & symptoms
        ELSE
            PROMPT user to upload kidney scan
    END
END

```

5.2 Results

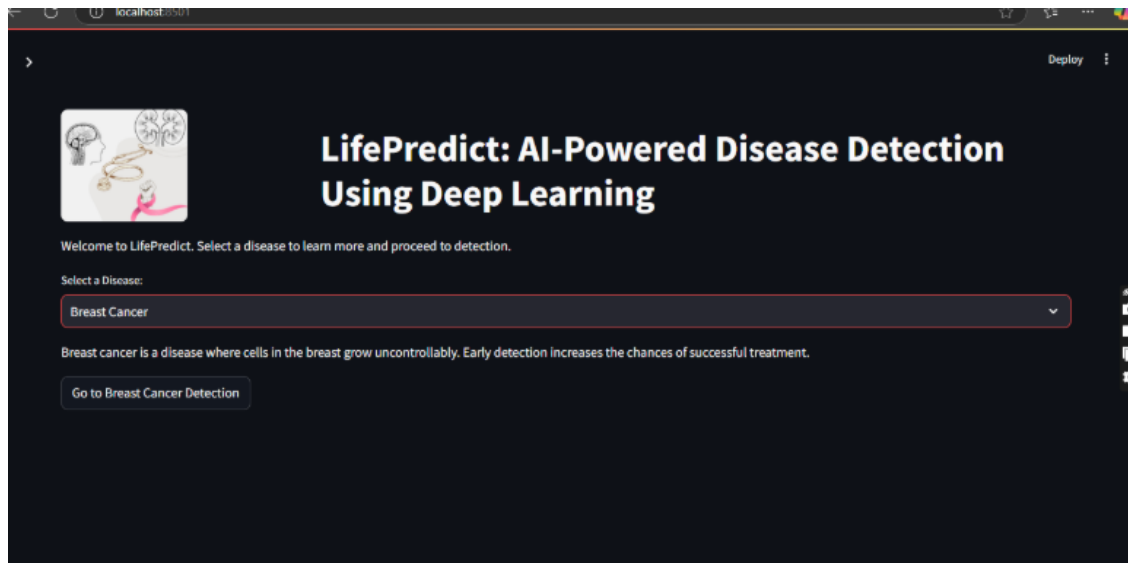


Figure 5.1: Selecting Breast Cancer For Detection

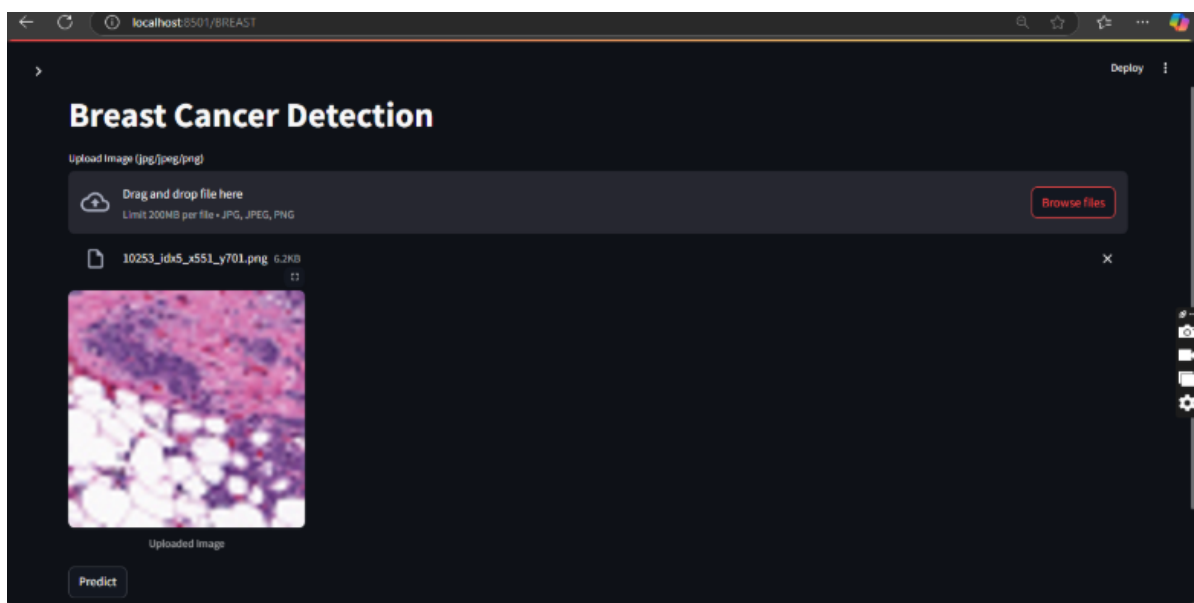


Figure 5.2: Upload The Image Of Breast Cancer

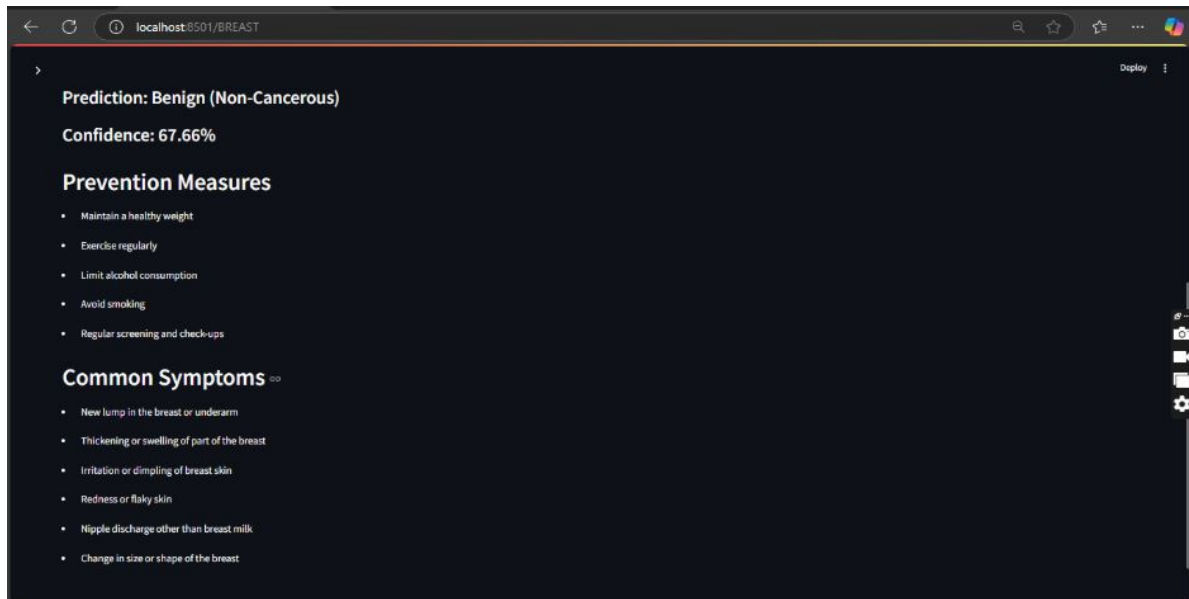


Figure 5.3: Result of Breast Cancer Uploaded Image

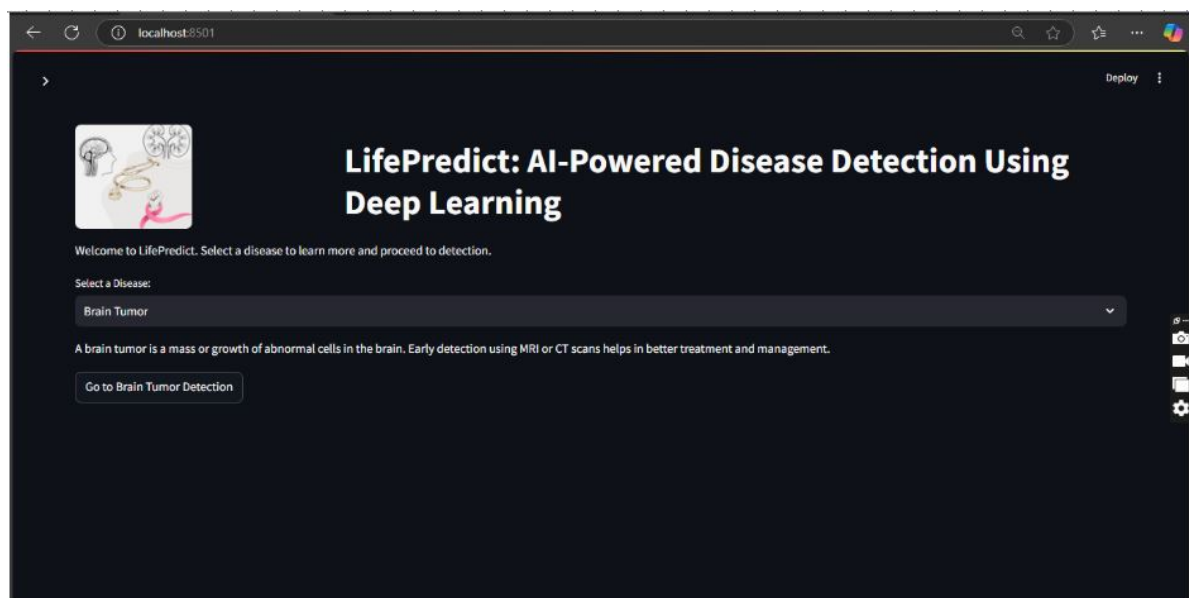


Figure 5.4: Selecting Brain Tumor For Detection

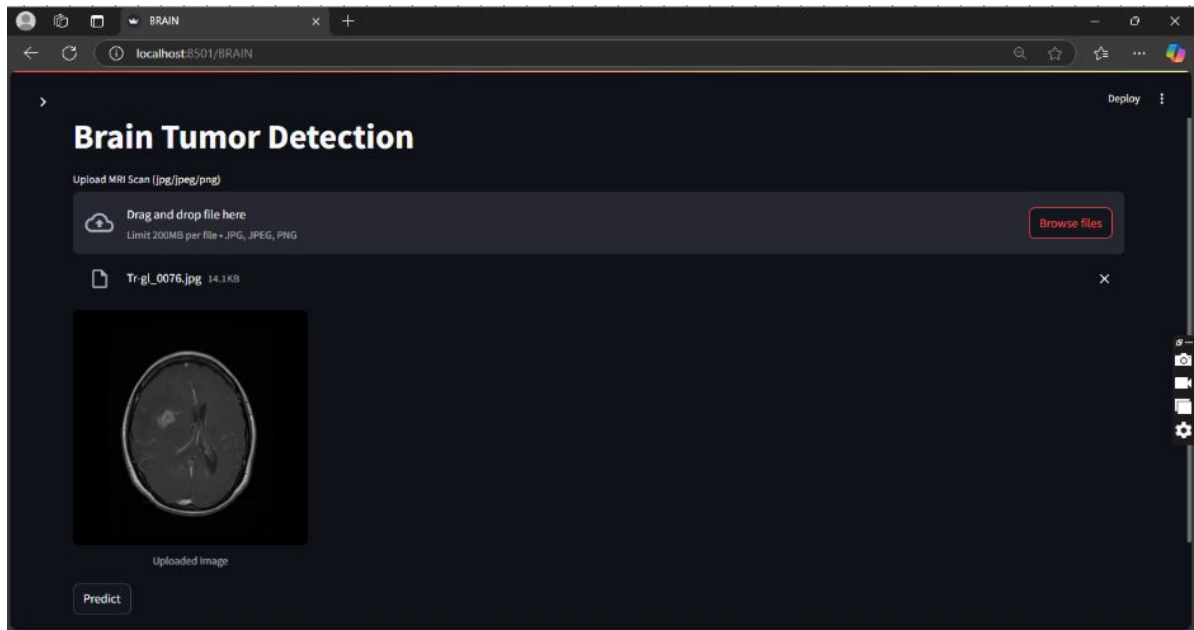


Figure 5.5: Upload The Image Of Brain Tumor

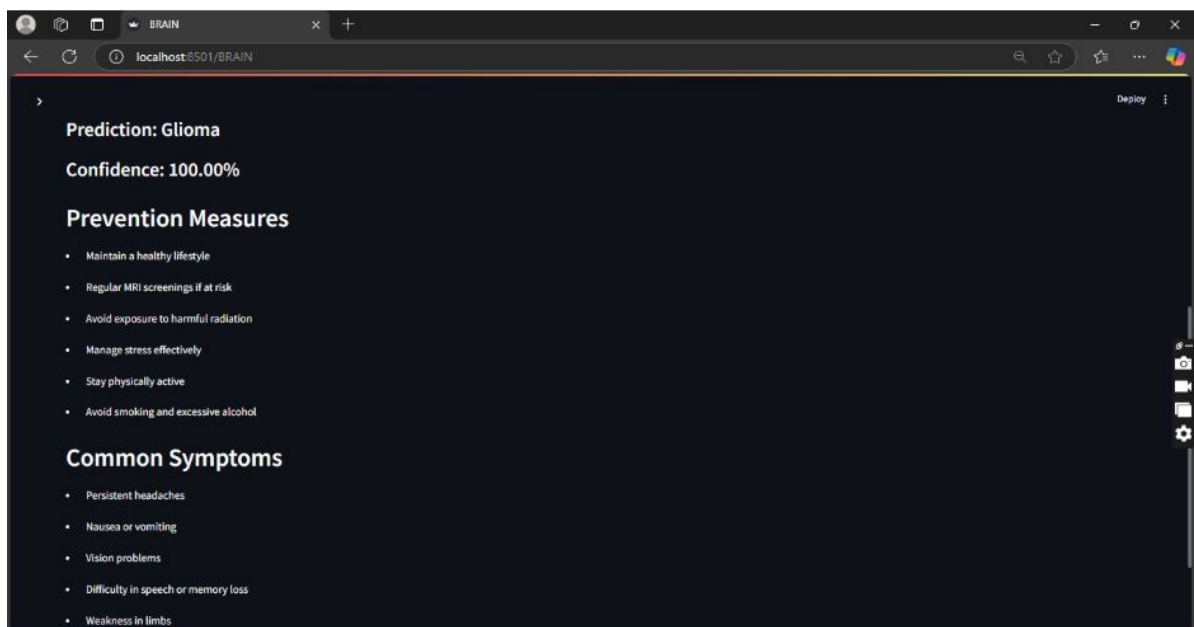


Figure 5.6: Result of Brain Tumor Uploaded Image

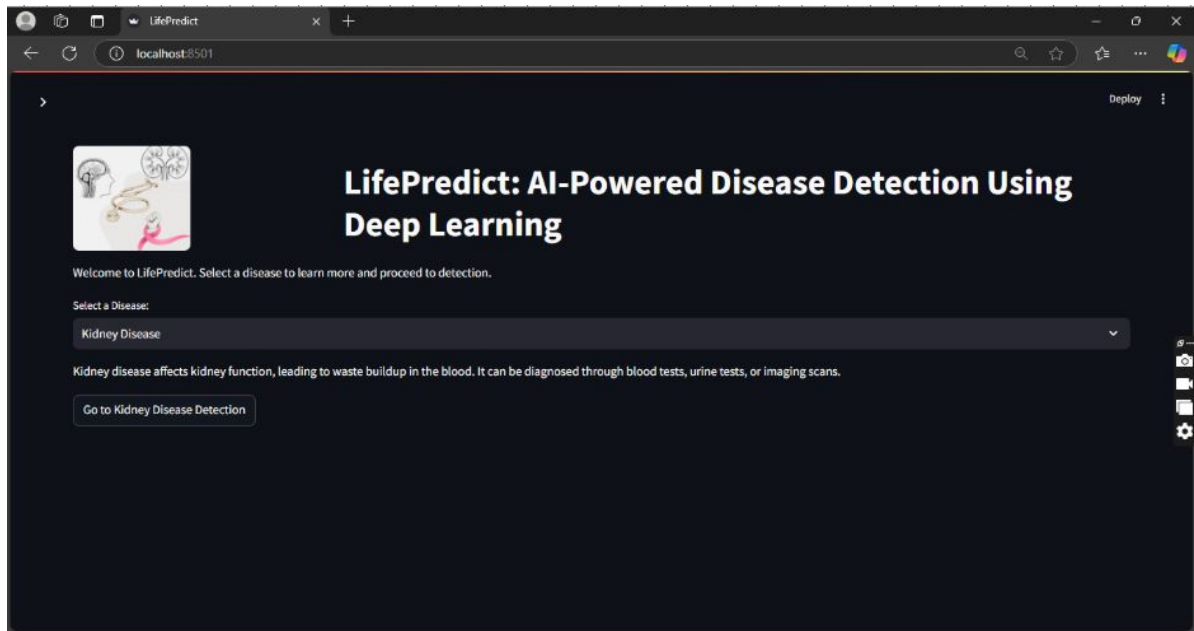


Figure 5.7: Selecting Kidney Stone For Detection

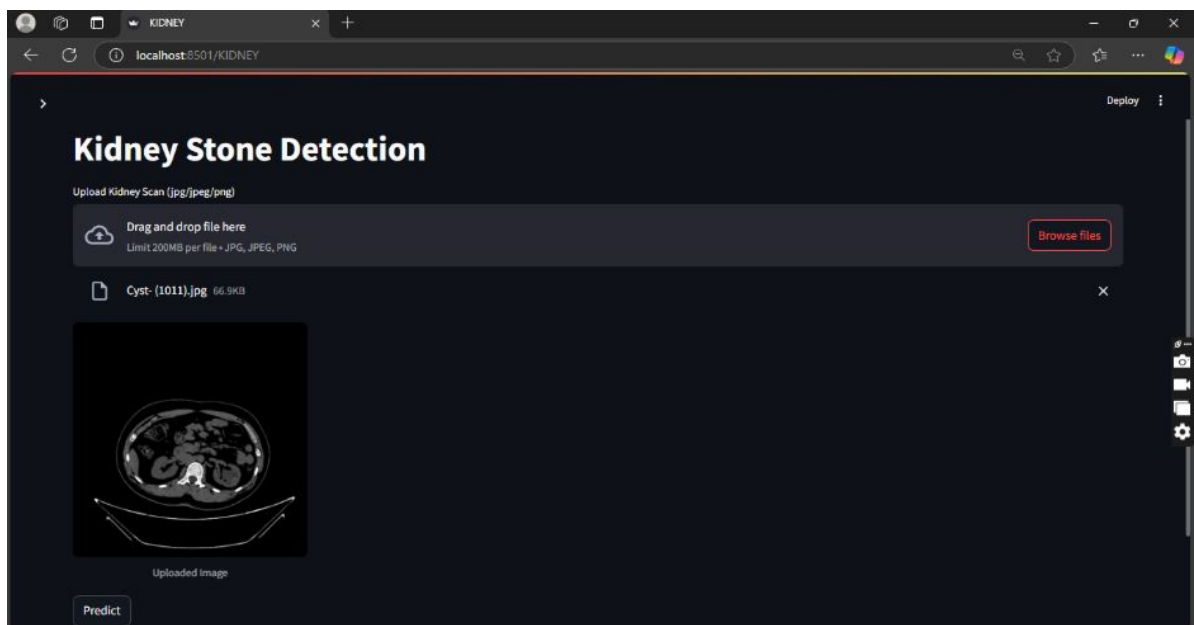


Figure 5.8: Upload The Image Of Kidney Stone

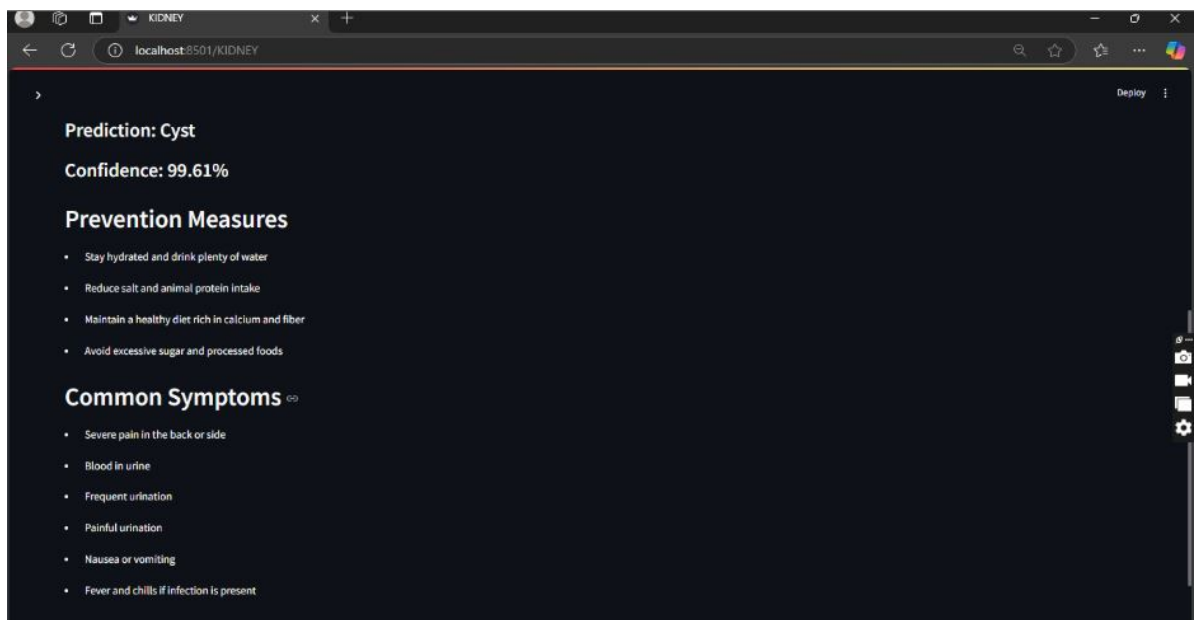


Figure 5.9: Result of Kidney Stone Uploaded Image

5.3 Discussions - Comparative Study

Paper	Accuracy (Brain Tumor)	Accuracy (Breast Cancer)	Accuracy (Kidney Stones/Tumor)	CNN Models Used	MCC Used	Multiclass Classification Testing
Our project	97%	87%	97%	Custom CNN, ResNet50, MobileNet	Yes	Yes
[4]	96%	NA	NA	CNN	No	No
[5]	NA	89.23%	NA	CNN, AlexNet, Stacked AutoEncoders	No	No
[9]	NA	NA	92%	CNN	No	No

Table 5.1: Comparison between Literature Survey and Existing System

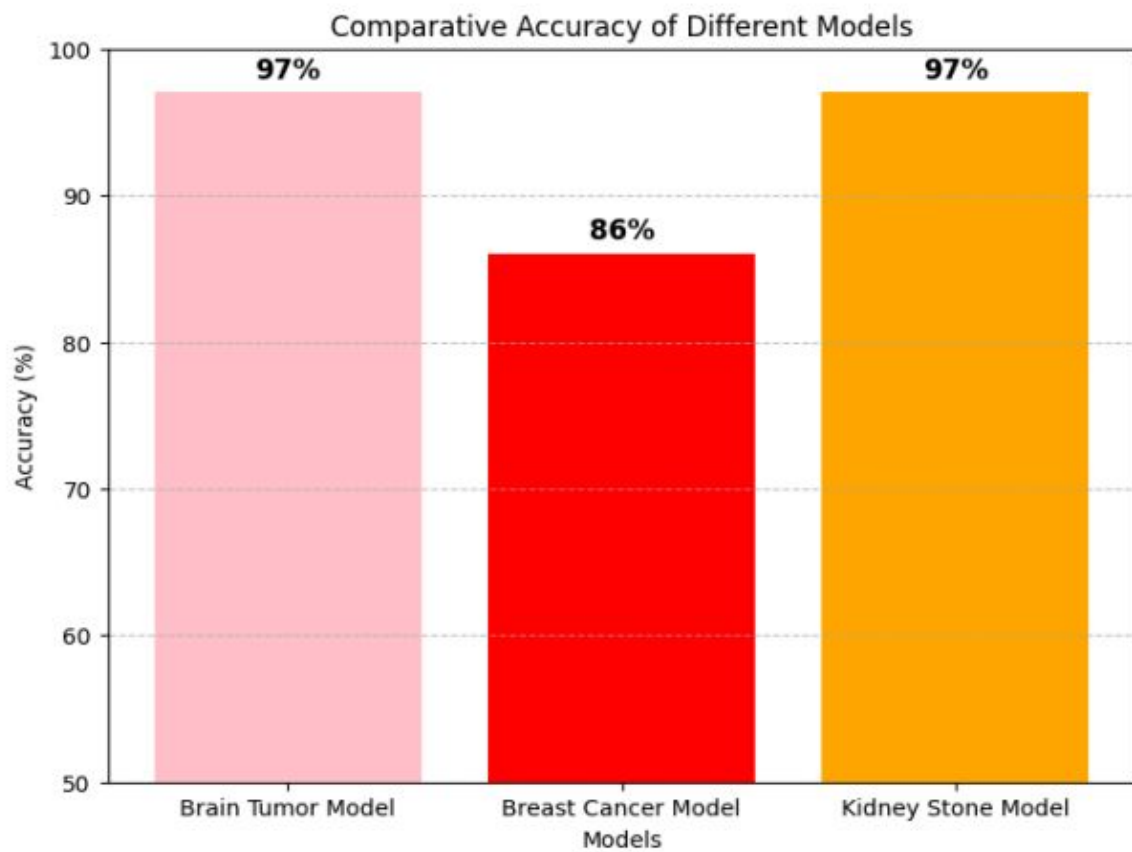


Figure 5.10: Proposed Model Accuracy Rate

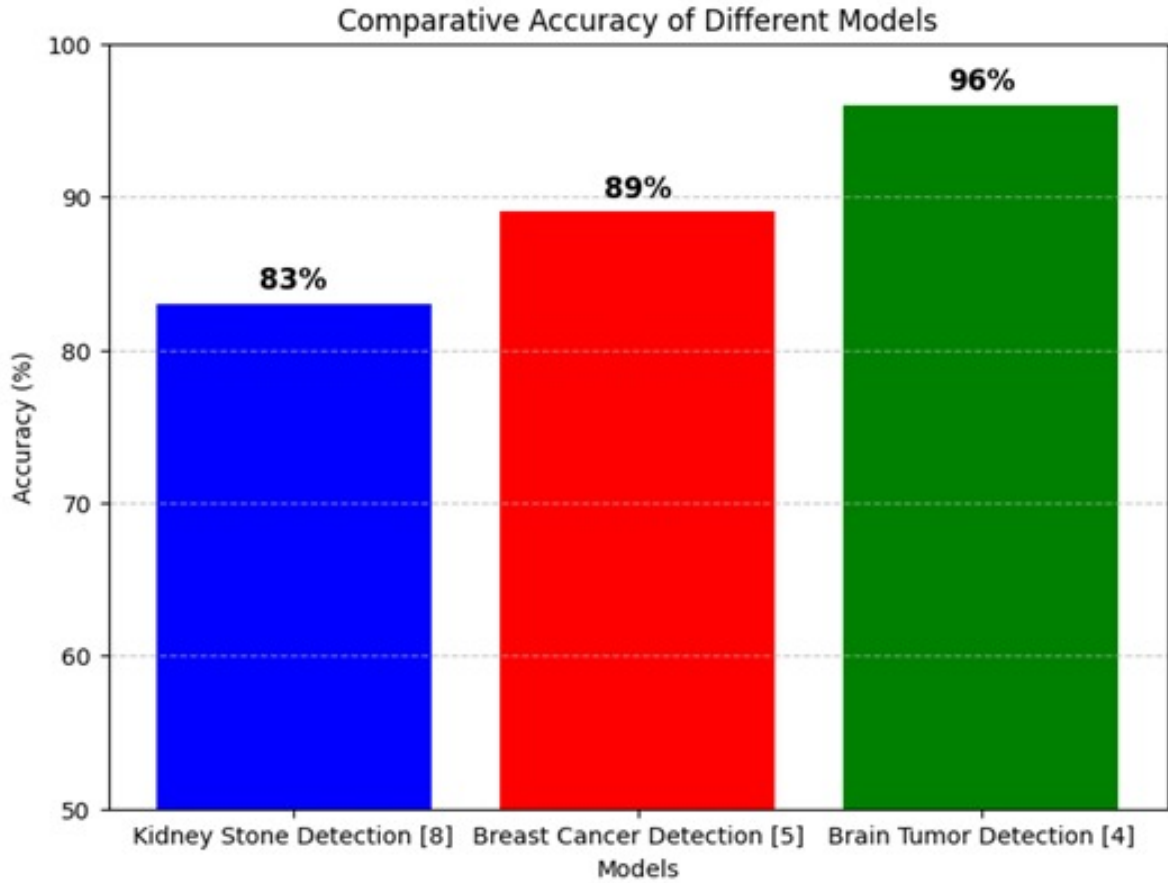


Figure 5.11: Comparative Paper Accuracy

The deep learning models developed for disease prediction in healthcare demonstrated promising results across three areas: brain tumor disease, kidney stone disease, and breast cancer disease. In brain tumor detection, two CNN models were implemented. The first model, a deep CNN with multiple convolutional blocks, achieved high accuracy in classifying MRI scans into four categories: gliomas, meningiomas, no tumors, and pituitary. The second, slightly shallower model, also performed well. Both models benefited from techniques such as batch normalization, dropout, and adaptive learning rate scheduling, which enhanced their generalization capabilities and reduced overfitting. In kidney stone classification, a dual approach was adopted. A transfer learning model based on ResNet50 was fine-tuned for the specific task, while a custom CNN architecture named MobileNet was also developed. Both models showed high accuracy in categorizing CT scan images into cyst, tumor, stone, and normal classes. The use of L2 regularization (also called weight decay) in MobileNet proved particularly effective in controlling overfitting. For breast cancer detection, the CNN model was designed to classify histopathological images. This model addressed the challenge of dataset imbalance through class weighting and data augmentation techniques. The implementation of three convolutional blocks with increasing filter numbers, along with strategic dropout layers, resulted in robust feature extraction and classification performance. Throughout all models, sophisticated methods of early stopping, model checkpointing, and learning rate decay strategies were applied to enhance training and avoid overfitting. Performance measures for the models included several metrics, including accuracy, precision,

recall, and F1-score, giving overall evaluation of their capacity, which is presented in Table 8. The results demonstrate the potential of deep learning in enhancing medical diagnostics. These models can serve as valuable tools for healthcare professionals, potentially improving the speed and accuracy of disease detection. The table provides a comparative overview of models, optimizers, learning rates, and parameter counts used for diagnosing three diseases: breast cancer, kidney stones, and brain tumors. For breast cancer detection, a CNN built with Keras uses the Adam optimizer and a relatively high learning rate of $1e-2$, yielding 186,692 total parameters and 185,474 trainable parameters. In the kidney stone domain, three different architectures are examined. The first is another Keras-based CNN, also optimized with Adam, but with a lower learning rate of $1e-3$; this configuration has 8,670,340 parameters in total, with 8,668,996 trainable. MobileNetV2, using a learning rate of 0.001, includes 3,432,412 parameters overall, but only 44,316 are trainable, indicating extensive parameter freezing or transfer learning. ResNet50, operating at an even lower learning rate of 0.0001, contains 24,683,582 parameters, with 4,134,784 of them trainable. For brain tumor classification, a VGGNet-based CNN is employed with a learning rate of 0.001, consisting of 2,613,636 total parameters and 2,610,756 trainable. These variations in parameter counts, learning rates, and trainable layers highlight the trade-off among model complexity, memory usage, and convergence behavior for different medical imaging tasks. The study stands out compared to other studies due to its higher accuracy across multiple diseases (97% brain tumor, 87% breast cancer, 97% kidney stones), whereas other papers either lack accuracy data for some diseases or report lower values. Unlike studies that use basic CNN models (e.g., AlexNet, standard CNN), research integrates Custom CNN, ResNet50, and MobileNet, enhancing feature extraction and classification performance. Additionally, MCC is exclusively used, providing a more reliable evaluation metric, while all other papers omit it. The study includes multiclass classification testing, whereas competing studies only focus on binary classification, making your research more clinically applicable and methodologically advanced.

Chapter 6

Conclusion & Future Work

6.1 Conclusion

In summary, the research points out the revolutionary application of deep learning in disease prognosis, especially in brain tumor, kidney stone, and breast cancer diagnosis. Using sophisticated methods like early stopping, model checkpoints, and learning rate reduction techniques, the models became highly accurate and resilient. They were thoroughly assessed using various parameters, which vindicated the authenticity and viability of deep learning-based medical diagnostics. These findings highlight the potential of deep learning to enhance the speed and accuracy of disease diagnosis, setting the stage for more effective and reliable diagnostic systems. Future work will concentrate on increasing datasets, improving model generalization, investigating interpretability techniques, and integrating these solutions into current healthcare workflows.

6.2 Future Work

Future work focuses on expanding the datasets to include a wider range of cases and variations, further improving the models' generalization capabilities. Additionally, explore interpretability techniques that enhance the trust and adoption of these models in clinical practice. The integration of these deep learning models into existing healthcare workflows also presents an exciting avenue for future research and development.

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Publication by Students

The paper "Enhancing Medical Imaging Interpretation With Deep Learning Models" is presented at *Fifth International Conference on Artificial Intelligence and Smart Computing (ICAISC-2025)* , Organized by Bannari Amman Institute of Technology.

Annexure

Enhancing Medical Imaging Interpretation With Deep Learning Models

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Abstract: The research introduces an advanced deep learning framework for medical imaging interpretation, significantly outperforming previous studies in disease classification. Unlike existing works that rely on traditional CNN models, the approach integrates optimized deep learning architectures, including ResNet50, MobileNet, and custom CNNs, achieving superior accuracy in detecting brain tumors, breast cancer, and kidney stones. A key distinction of a work is the incorporation of Matthews Correlation Coefficient (MCC) and rigorous multiclass classification testing, which provide a more balanced evaluation, especially for imbalanced datasets—a limitation often overlooked in prior research. The models achieve 97% accuracy in brain tumor classification, 87% in breast cancer detection, and 97% in kidney stone classification, demonstrating enhanced generalization across diverse medical datasets. Compared to previous studies, which primarily use standard CNNs without comprehensive performance validation, the research ensures higher precision and reliability through advanced feature extraction, optimized training strategies, and robust evaluation metrics. Furthermore, by deploying our models in a web-based diagnostic platform, we bridge the gap between deep learning advancements and real-world clinical applications, offering an efficient and accessible solution for automated disease detection.

Keywords: deep learning, disease prediction, convolutional neural network (CNN), kidney stone, brain tumor, breast cancer.

I. INTRODUCTION

Medical imaging is key to the early diagnosis and detection of lethal diseases. Conventional diagnostic processes depend on manual inspection of medical scans, which may take time and is susceptible to human error. The incorporation of deep learning (DL) into healthcare has brought about new avenues of automating and enhancing disease detection, increasing diagnostic accuracy and efficacy. In the research, we present a web-based diagnostic platform that employs deep learning models for detecting breast cancer, brain tumors, and kidney stones using medical imaging.

Breast cancer is among the most common and lethal conditions in women globally, whereby early diagnosis with histopathology imaging greatly enhances survival. Histopathology images present microscopic images of tissue specimens, which facilitate proper identification of cancerous cells[5]. Our breast cancer detection model is trained on such images and classifies them into two categories: benign (non-cancerous) and malignant (cancerous). Similarly, brain tumors require early diagnosis for effective treatment. Using Magnetic Resonance Imaging (MRI) scans, the model

categorizes brain tumors into four distinct classes—glioma, no tumor, meningioma, and pituitary—with studies demonstrating high accuracies in this domain [3]. Additionally, kidney stone disease (nephrolithiasis) is a common urological condition that, if untreated, can lead to severe complications. Diagnosis is traditionally performed using X-rays, ultrasound, MRI, and Computed Tomography (CT) scans, all of which require manual interpretation by radiologists. Our deep learning model automates kidney stone detection and classifies medical images into four categories: cyst, tumor, stone, and normal [1].

To develop the system, we curated and preprocessed three distinct medical image datasets, ensuring robust feature extraction. We employed Convolutional Neural Networks (CNNs) with multiple layers to learn deep representations from these images and achieve high classification accuracy, with notable accuracies such as 84% and 97% for brain tumor detection [4] demonstrating their effectiveness in real-world applications. Furthermore, confusion matrices were analyzed to evaluate model performance, ensuring reliability for clinical use. The prime directive of this research is to develop a scalable, effortless, and fully automated diagnostic system that leverages CNNs to process medical images and generate real-time

predictions. By integrating advanced deep learning architectures with web-based deployment, the platform ensures efficient feature extraction, high-throughput image analysis, and robust classification performance. Additionally, the study examines key challenges and potential advancements in deep learning-driven diagnostic frameworks, focusing on dataset diversity, model robustness across varying imaging conditions, and seamless integration into clinical workflows.

II. LITERATURE SURVEY

A literature survey helps in understanding recent advancements in a field of study and plays a crucial role in shaping research by identifying gaps in existing studies. It provides an outline of performed activities by various authors and researchers. The section analyzes research related to the application of deep learning. Reviewing existing literature helps in understanding key concepts, exploring previous findings, and identifying areas for improvement. It also connects new research with existing data, ensuring a strong foundation for further advancements.

[1] reviewed different ML and DL methods for kidney stone detection, such as Support Vector Machines (SVM), Random Forest, Decision Trees, and Convolutional Neural Networks (CNN). The research, however, did not have experimental verification and did not introduce a new model.

[2] proposed CNN-6, ResNet50, and VGG16-based models for kidney tumor detection, achieving accuracies of 97%, 96%, and 92%, respectively. The study emphasized the need for larger datasets to make more versatile and practically useful models.

In [3], implementation of CNN, VGG-16, and ensemble learning approaches is described for MRI-based classification, achieving 98.14% accuracy. Despite its high performance, the study highlighted computational constraints and extended training times.

Similarly, [4] applied CNN for tumor detection in MRI scans, achieving 96% accuracy. However, the model was limited to binary classification (Tumor/No Tumor) and did not differentiate between tumor subtypes.

[5], with DCNN achieving the highest accuracy of 89.23%. The study noted performance degradation due to class imbalance between benign and malignant cases.

[6] compared ANN, CNN, SVM, Random Forest, KNN, and Logistic regression systems for forecasting breast cancer. ANN demonstrated the highest accuracy of 99.3%, followed by CNN (97.3%) and SVM with Random Forest (96.5%). However, the study was conducted on structured datasets rather than real-world histopathology images, limiting its clinical applicability.

[7] analyzes multiple deep learning models, including CNN, SVM, KNN, Naïve Bayes, and Decision Trees, for breast cancer detection. While the study provided valuable insights, it lacked real-world implementation and direct performance comparisons.

[8] has described the implementation of CNN-based deep learning with transfer learning, achieving accuracy of 83.2% (surface), 90.4% (section), and 85.6% (mixed). The research faced challenges related to small datasets, class imbalance, and dependence on ImageNet for feature extraction.

Furthermore, [9] investigated kidney stone detection using CNN, GLCM feature extraction, Fuzzy C-means, and back propagation Neural Networks (BPN), reporting an accuracy of 92.57%. However, the study encountered challenges such as noise in ultrasound images and a limited dataset.

[10] conducted a study comparing deep learning models such as CNN, VGG16, VGG19, and InceptionV3 for brain tumor detection using medical images. Their research found that VGG16 obtained the best accuracy of 98%, demonstrating its efficiency in tumor classification. Yet they mentioned that no data augmentation measures, including rotation or cropping, were used, which may affect model generalization.

These studies collectively highlight the advancements and DL for medical diagnostics while also emphasizing these challenges, which include dataset restrictiveness, imbalance classes, increased computation expense, and the requirement of real-world testing. It should be future work to advance dataset diversity and enhance deep learning models for medical deployment in clinical processes.

III. METHODOLOGY

A. DATA COLLECTION

Developing accurate deep learning (DL) models for medical diagnostics requires high-quality datasets. Medical imaging datasets, including CT scans, MRIs, and X-rays, are widely used for detecting diseases such as brain tumors, kidney stones, and breast cancer. However, a major challenge in this domain is the scarcity of labeled data, particularly for intricate scenarios. In response to this problem, dataset augmentation

Methods like flipping and scaling are used to improve the performance of models and enhance generalization.

A dataset of 7,023 MRI images was used for brain tumor detection, classified into glioma, meningioma, pituitary tumors, and no tumor. These MRI images were carefully labeled and verified by medical professionals to ensure accuracy. Similarly, in breast cancer detection, a dataset containing 277,524 50x50 image patches was created to identify Invasive Ductal Carcinoma (IDC). The dataset was labeled as malignant or benign, aiding in the automatic classification and grading of IDC regions to assess aggressiveness. For kidney stone detection, a dataset of 12,446 CT scan images was collected from hospitals in Dhaka, Bangladesh. These images were carefully labeled into four categories: cyst, tumor, stone, and normal, following expert verification by radiologists. The inclusion of verified medical datasets, combined with augmentation techniques, plays a crucial role in improving model accuracy, enabling early detection, and supporting clinical decision-making for better patient care.

B. DATA SPLITTING

The Kidney Stone, Breast Cancer, and Brain Tumor datasets were split into 80% training, 10% validation, and 10% testing for a balanced evaluation.

For the Kidney Stone Dataset, images were categorized into Cyst, Tumor, Stone, and Normal. The split ensured equal class representation, with 2,967 cyst, 1,826 tumor, 1,101 stone, and 4,061 normal images for training, while the remaining were divided for validation and testing shown in below fig 1.

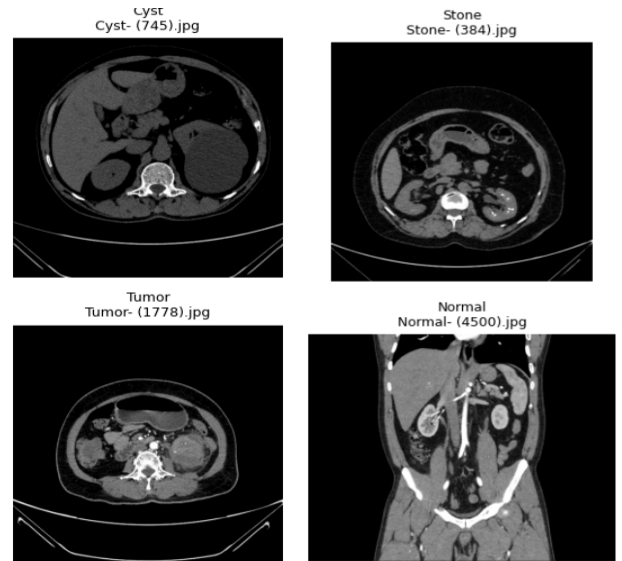


Fig. 1: Sample Images (Top Left: Cyst (Class 0), Bottom Right: Normal (Class 1), Top Right: Stone (Class 2) and Bottom Left: Tumor (Class 3))

The Brain Tumor Dataset was initially split 80% for training and 20% for testing, with further refinement into 90% training and 10% validation. The training set included 1,457 pituitary, 1,595 no-tumor, 1,339 meningioma, and 1,321 glioma images, while testing had 300 pituitary, 405 no-tumor, 306 meningioma, and 300 glioma images. Where classes are shown below in fig 2

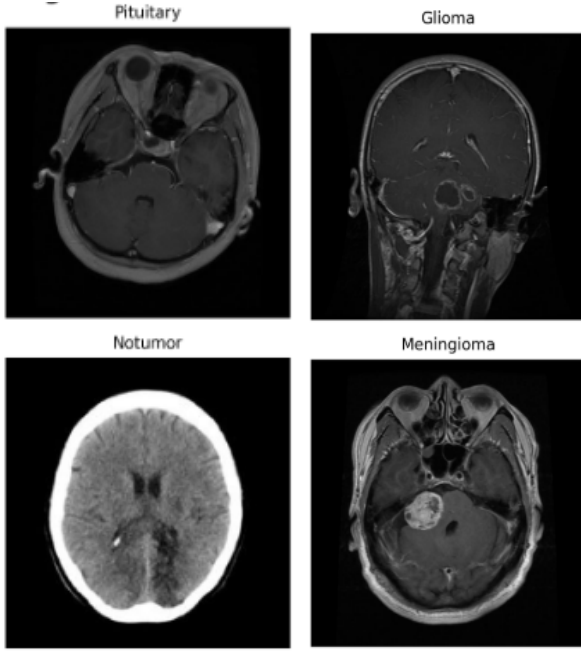


Fig. 2: Sample Images (Top Left: Pituitary (Class 0), Bottom Right: Meningioma (Class 1), Top Right: Glioma (Class 2) and Bottom Left: No Tumor (Class 3))

For the Breast Cancer Dataset, the training set included 72,633 malignant and 183,361 benign images; the validation set had 12,195 malignant and 30,525 benign images; and the testing set contained 28,518 malignant and 71,563 benign images.

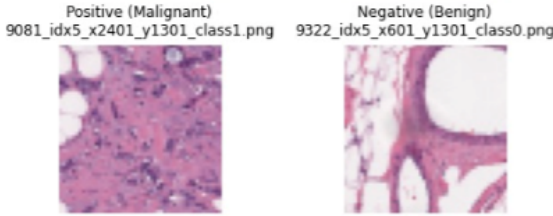


Fig. 3: Sample Images Left: Malignant and Right: Benign

C. DATA PROCESSING AND AUGMENTATION

Effective data preprocessing enhances model performance and generalization in deep learning. For the kidney stone dataset, images were rescaled to $[0, 1]$, resized to 64×64 pixels, and divided into 80% training, 10% validation, and 10% testing. Data augmentation for training included shearing (0.2), zooming (0.2), and horizontal flipping, while validation and testing sets were only rescaled. Images were categorized into subdirectories for loading.

For the Breast Cancer Dataset, similar preprocessing was applied. Images were rescaled, resized to 48×48 pixels, and split (80% training, 10% validation, 10% testing). Augmentation included rotation (20°), zooming (5%), shifting (10%), shearing (0.05), and flipping.

In the Brain Tumor Dataset, images were rescaled, resized to 150×150 pixels, and split identically. Augmentation included rotation (20°), zooming (5%), shifting (10%), shearing (0.05), and flipping, with validation/testing sets only rescaled. Images were stored in labeled subdirectories for efficient loading.

The data augmentation parameters are set up to enhance diversity in the training data. The images are scaled by a factor of $1/255$ during pixel

value normalization between 0 and 1. Also, random rotations up to 20 degrees are added to mimic varied orientations. The images are also randomly zoomed with a range of 0.05, and horizontal and vertical shifts of up to 10% of the image size are added. A shear transformation with a range of 0.05 also distorts the images to simulate real-world variations. Horizontal and vertical flip are both enabled to add to the dataset's diversity, and the 'nearest' fill mode is employed to complete any gaps resulting from these transformations. These preprocessing and augmentation steps ensure robust medical image classification by improving model accuracy and generalization.

D. MODEL TRAINING

Developed multiple deep learning models aimed at medical image classification, specifically for detecting brain tumors, kidney stones, and breast cancer. Each of these models has been carefully designed using convolutional neural networks (CNNs), incorporating advanced techniques such as batch normalization, dropout, and regularization to enhance accuracy while minimizing the risk of overfitting. Approach involves using both custom-built CNN architectures and transfer learning with pre-trained models, ensuring that the models generalize well on unseen data. Additionally, implemented strategies such as class weighting to handle dataset imbalance, data augmentation to improve robustness, and various callback functions to optimize training performance.

In brain tumor classification, there are two models designed. The initial model is a CNN with a series of convolutional blocks having an accuracy of 97%, each preceded by batch normalization to regularize learning and max pooling to diminish spatial dimensions without sacrificing significant features. The filters in the convolutional layers increase successively from 32 to 512 so that the model can detect both low-level and high-level features in the brain MRI scans. Instead of using traditional flattening, employ a GlobalAveragePooling2D layer, which reduces overfitting by averaging feature maps instead of merely reshaping them.

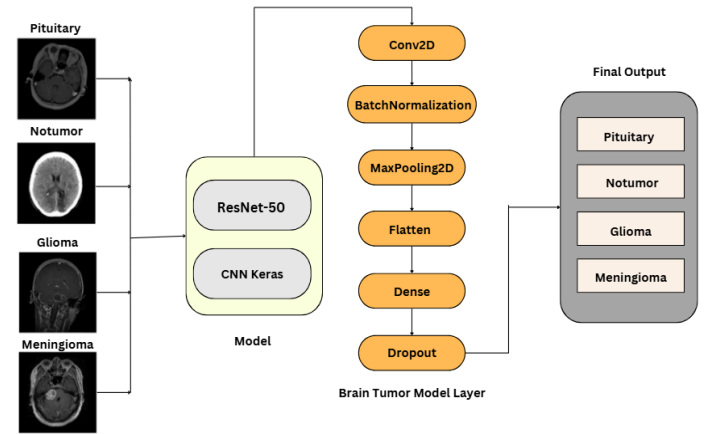


Fig. 4: Brain Tumor Model

The fully connected layers that follow contain dropout to further enhance generalization, and the final softmax layer provides classification into one of the four brain tumor categories. Additionally, multiple callbacks, including EarlyStopping, which stops training if validation loss does not improve anymore; ModelCheckpoint, which stores the best model during training; and ReduceLROnPlateau, which reduces the learning rate if the model does not improve anymore, resulting in better convergence. The second brain tumor classification model follows a similar architecture but is slightly shallower, with fewer convolutional filters and layers, using a Flatten() layer instead of GlobalAveragePooling2D shown in below table 1.

Layer (type)	Output Shape	Param #
conv2d_8 (Conv2D)	(None, 48, 48, 32)	896
batch_normalization_1 (BatchNormalization)	(None, 48, 48, 32)	128
conv2d_9 (Conv2D)	(None, 48, 48, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 48, 48, 64)	256
max_pooling2d_3 (MaxPooling2D)	(None, 24, 24, 64)	0
dropout_5 (Dropout)	(None, 24, 24, 64)	0
conv2d_10 (Conv2D)	(None, 24, 24, 64)	36,928
batch_normalization_2 (BatchNormalization)	(None, 24, 24, 64)	256
conv2d_11 (Conv2D)	(None, 24, 24, 128)	73,856
batch_normalization_3 (BatchNormalization)	(None, 24, 24, 128)	512
max_pooling2d_4 (MaxPooling2D)	(None, 12, 12, 128)	0
dropout_6 (Dropout)	(None, 12, 12, 128)	0
conv2d_12 (Conv2D)	(None, 12, 12, 128)	147,584
batch_normalization_4 (BatchNormalization)	(None, 12, 12, 128)	512
conv2d_13 (Conv2D)	(None, 12, 12, 256)	295,168
batch_normalization_5 (BatchNormalization)	(None, 12, 12, 256)	1,024
max_pooling2d_5 (MaxPooling2D)	(None, 6, 6, 256)	0
dropout_7 (Dropout)	(None, 6, 6, 256)	0
conv2d_14 (Conv2D)	(None, 6, 6, 256)	590,080
batch_normalization_6 (BatchNormalization)	(None, 6, 6, 256)	1,024
conv2d_15 (Conv2D)	(None, 6, 6, 512)	1,180,160
batch_normalization_7 (BatchNormalization)	(None, 6, 6, 512)	2,048
dropout_8 (Dropout)	(None, 6, 6, 512)	0
global_average_pooling2d (GlobalAveragePooling2D)	(None, 512)	0
dense_2 (Dense)	(None, 512)	262,656
dropout_9 (Dropout)	(None, 512)	0
dense_3 (Dense)	(None, 4)	2,052

Table 1: Summary of Brain Tumor

max pooling. In contrast to deeper models, KidneyNet is comparatively shallow but attains high accuracy through the use of L2 regularization, which prevents overfitting by discouraging large weight values in all layers. The final softmax layer ensures precise classification by outputting probabilities for the four kidney stone categories. Summarize of kidney stones is given in Table 2.

Layer (type)	Output Shape	Param #	Connected to
input_layer_1 (InputLayer)	(None, 224, 224, 3)	0	-
conv1_pad (ZeroPadding2D)	(None, 230, 230, 3)	0	input_layer_1[0][0]
conv1_conv (Conv2D)	(None, 112, 112, 64)	9,472	conv1_pad[0][0]
conv1_bn (BatchNormalization)	(None, 112, 112, 64)	256	conv1_conv[0][0]
conv1_relu (Activation)	(None, 112, 112, 64)	0	conv1_bn[0][0]
pool1_pad (ZeroPadding2D)	(None, 114, 114, 64)	0	conv1_relu[0][0]
pool1_pool (MaxPooling2D)	(None, 56, 56, 64)	0	pool1_pad[0][0]
conv2_block1_1_conv (Conv2D)	(None, 56, 56, 64)	4,160	pool1_pool[0][0]
conv2_block1_1_bn (BatchNormalization)	(None, 56, 56, 64)	256	conv2_block1_1_conv[0..
conv2_block1_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_1_bn[0][..
conv2_block1_2_conv (Conv2D)	(None, 56, 56, 64)	36,928	conv2_block1_1_relu[0..
conv2_block1_2_bn (BatchNormalization)	(None, 56, 56, 64)	256	conv2_block1_2_conv[0..
conv2_block1_2_relu (Activation)	(None, 56, 56, 64)	0	conv2_block1_2_bn[0][..
conv2_block1_0_conv (Conv2D)	(None, 56, 56, 256)	16,640	pool1_pool[0][0]
conv2_block1_3_conv (Conv2D)	(None, 56, 56, 256)	16,640	conv2_block1_2_relu[0..
conv2_block1_0_bn (BatchNormalization)	(None, 56, 56, 256)	1,024	conv2_block1_0_conv[0..
conv2_block1_3_bn (BatchNormalization)	(None, 56, 56, 256)	1,024	conv2_block1_3_conv[0..
conv2_block1_add (Add)	(None, 56, 56, 256)	0	conv2_block1_0_bn[0][.. conv2_block1_3_bn[0][..
conv2_block1_out (Activation)	(None, 56, 56, 256)	0	conv2_block1_add[0][0]
conv2_block2_1_conv (Conv2D)	(None, 56, 56, 64)	16,448	conv2_block1_out[0][0]
conv2_block2_1_bn (BatchNormalization)	(None, 56, 56, 64)	256	conv2_block2_1_conv[0..
conv2_block2_1_relu (Activation)	(None, 56, 56, 64)	0	conv2_block2_1_bn[0][..

conv5_block3_1_conv (Conv2D)	(None, 7, 7, 512)	1,049,088	conv5_block2_out[0][0]
conv5_block3_1_bn (BatchNormalization)	(None, 7, 7, 512)	2,048	conv5_block3_1_conv[0..
conv5_block3_1_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_1_bn[0][..
conv5_block3_2_conv (Conv2D)	(None, 7, 7, 512)	2,359,808	conv5_block3_1_relu[0..
conv5_block3_2_bn (BatchNormalization)	(None, 7, 7, 512)	2,048	conv5_block3_2_conv[0..
conv5_block3_2_relu (Activation)	(None, 7, 7, 512)	0	conv5_block3_2_bn[0][..
conv5_block3_3_conv (Conv2D)	(None, 7, 7, 2048)	1,050,624	conv5_block3_2_relu[0..
conv5_block3_3_bn (BatchNormalization)	(None, 7, 7, 2048)	8,192	conv5_block3_3_conv[0..
conv5_block3_add (Add)	(None, 7, 7, 2048)	0	conv5_block2_out[0][0.. conv5_block3_3_bn[0][..
conv5_block3_out (Activation)	(None, 7, 7, 2048)	0	conv5_block3_add[0][0]
global_average_pooling2d... (GlobalAveragePooling2D)	(None, 2048)	0	conv5_block3_out[0][0]
dense_2 (Dense)	(None, 512)	1,049,088	global_average_poolin...
dropout_1 (Dropout)	(None, 512)	0	dense_2[0][0]
dense_3 (Dense)	(None, 4)	2,052	dropout_1[0][0]

Table 2: Summary of Kidney Stone

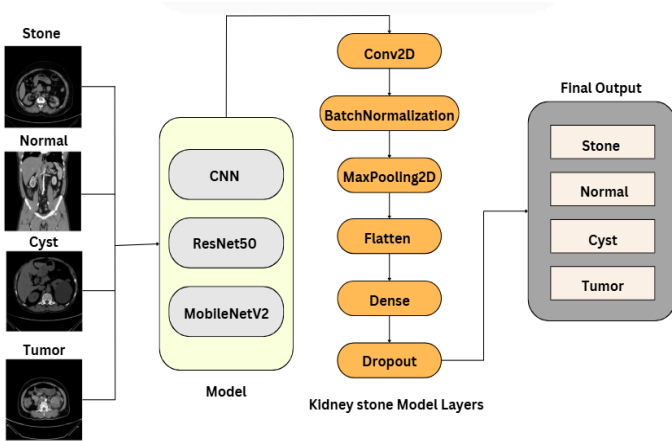


Fig. 5: Kidney Stone Model

In the breast cancer classification, a CNN model using keras is specifically designed for classifying histopathological images of breast cancer. The architecture uses three convolutional blocks with an increasing number of filters in order to extract useful features from the images. Batch normalization is used to stabilize learning, and dropout layers with values of 0.25 to 0.5 are appropriately placed in order to avoid overfitting. Unlike

traditional dense networks, the model ensures that the extracted features are well-processed before reaching the final fully connected layers, which refine the classification before passing through the softmax output layer. One major challenge in breast cancer detection is dataset imbalance, where cancerous samples are less frequent than non-cancerous ones. For this purpose, class weight implementation is done, with greater weights to underrepresented classes so that the model does not get biased against the majority class. Data augmentation methods such as rotation, shifting width and height, zoom, and flip to artificially enhance dataset diversity and make the model generalize and summarize better for breast cancer are presented in Table 3.

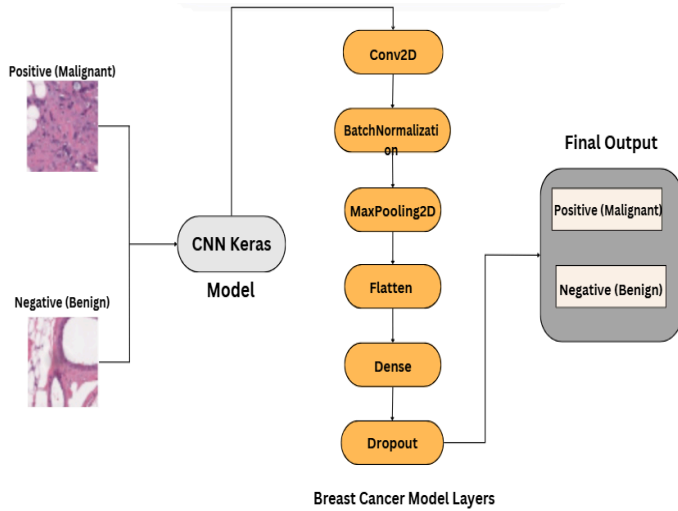


Fig. 6: Breast Cancer Model

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 46, 46, 32)	896
batch_normalization (BatchNormalization)	(None, 46, 46, 32)	184
max_pooling2d (MaxPooling2D)	(None, 23, 23, 32)	0
dropout (Dropout)	(None, 23, 23, 32)	0
conv2d_1 (Conv2D)	(None, 21, 21, 64)	18,496
batch_normalization_1 (BatchNormalization)	(None, 21, 21, 64)	84
conv2d_2 (Conv2D)	(None, 19, 19, 64)	36,928
batch_normalization_2 (BatchNormalization)	(None, 19, 19, 64)	76
max_pooling2d_1 (MaxPooling2D)	(None, 9, 9, 64)	0
dropout_1 (Dropout)	(None, 9, 9, 64)	0
conv2d_3 (Conv2D)	(None, 7, 7, 64)	36,928
batch_normalization_3 (BatchNormalization)	(None, 7, 7, 64)	28
conv2d_4 (Conv2D)	(None, 5, 5, 64)	36,928
batch_normalization_4 (BatchNormalization)	(None, 5, 5, 64)	20
conv2d_5 (Conv2D)	(None, 3, 3, 64)	36,928
batch_normalization_5 (BatchNormalization)	(None, 3, 3, 64)	12
max_pooling2d_2 (MaxPooling2D)	(None, 1, 1, 64)	0
dropout_2 (Dropout)	(None, 1, 1, 64)	0
flatten (Flatten)	(None, 64)	0
dense (Dense)	(None, 256)	16,640
batch_normalization_6 (BatchNormalization)	(None, 256)	1,024
dropout_3 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 2)	514

Table 3: Summary of Breast Cancer Model

Adoption of the best, optimal models is done to ensure robust training and evaluation. To counteract overfitting, apply dropout layers, L2 regularization, and global average pooling, which minimize the number of trainable parameters and avoid the model memorizing the training data rather than learning informative features. Implementation of learning rate reduction strategies, ensuring that the optimizer gradually fine-tunes weights for optimal performance. Training was monitored using validation loss and accuracy, and models were tested using unseen data to ensure real-world applicability. Moreover, advanced techniques such as early stopping and model checkpoints save the best-performing model automatically during training, preventing unnecessary overfitting and long training times.

Effectively combines custom CNN architectures with transfer learning, making it highly effective for medical image classification tasks. By incorporating multiple best practices, such as data augmentation, class balancing, dropout, and adaptive learning rate scheduling, it is ensured that models perform optimally on complex datasets. Models have the potential to significantly aid in the early detection of brain tumors, kidney stones, and breast cancer, ultimately contributing to better diagnostic outcomes in medical imaging.

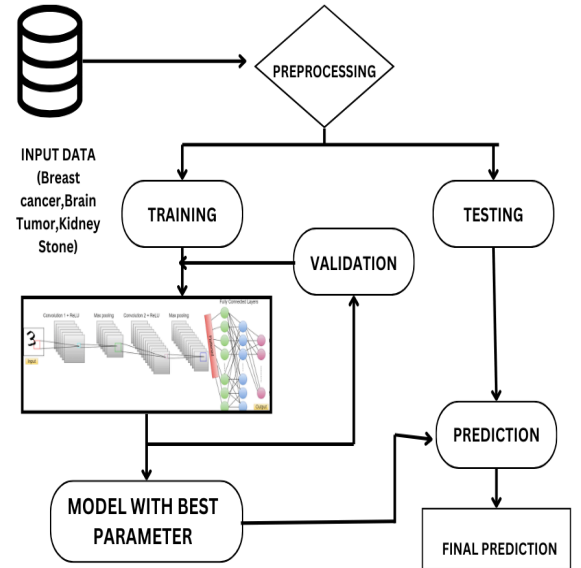


Fig. 7: The flow of the CNN Model

E. PERFORMANCE EVALUATION

Several performance measures are used regularly to contrast the performance of a system. They are Accuracy, Precision, Recall, AUC (Area Under the Curve), F1-Score, the Confusion Matrix, Matthews Correlation coefficient (MCC) and the Receiver Operating Characteristic (ROC) Curve. Each measure indicates a different aspect of model performance, and their definitions, as well as their mathematical representations, are explained in detail below.

• Confusion Matrix

A confusion matrix is a two-dimensional table that summarizes the result of the predictions performed by the classification performed by and allows to compare the correct and incorrect results of the prediction, which allows to visualize at which step a model can be confused in its predictions and to measure these performances. Within a confusing matrix, the outcomes are put into four major groups: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). The four components of the confusion matrix are presented in the tables below.

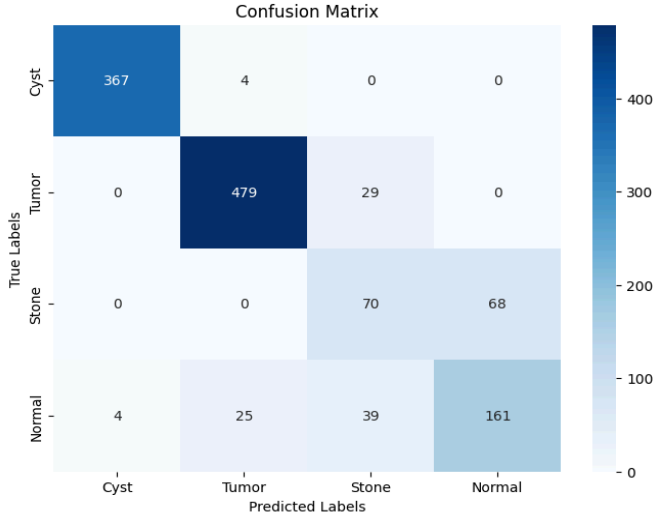


Fig 8: Kidney Stone Confusion matrix

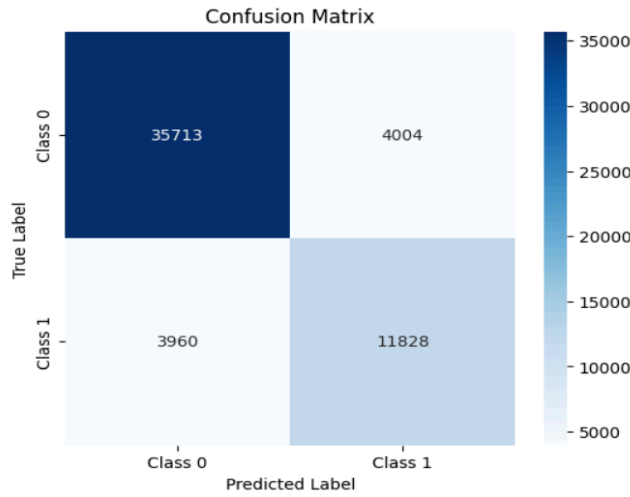


Fig 9: Breast Cancer Confusion Matrix

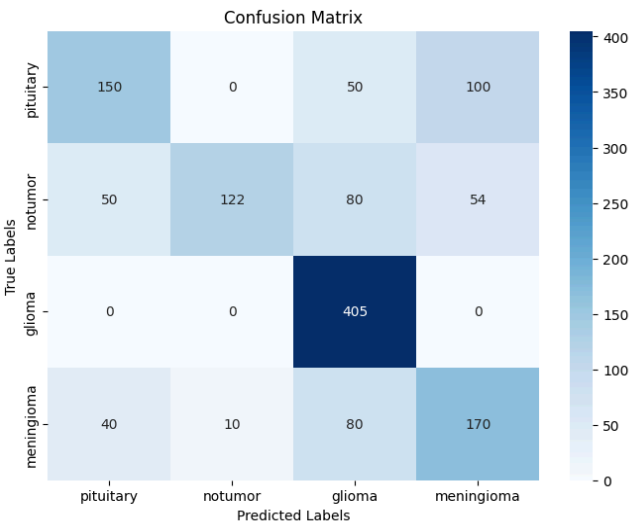


Fig 10 : Brain Tumor Confusion Matrix

- **Accuracy**

Accuracy computes the number of instances that were correctly classified out of total instances. Accuracy is useful if the classes are well balanced but can be misleading for imbalanced datasets.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \quad \text{Eq. 1. Accuracy}$$

- **Precision (Positive Predictive Value)**

Accuracy computes the number of correctly predicted positive instances out of all the instances predicted positive. It is useful in scenarios where false positives are costly.

$$Precision = \frac{TP}{TP + FP} \quad \text{Eq. 2. Precision}$$

- **Recall**

Recall is the proportion of positive cases that are correctly identified. It is worth it when the loss of a positive case is costly (e.g., medical diagnosis).

$$Recall = \frac{TP}{TP + FN} \quad \text{Eq. 3. Recall}$$

- **F1-Score**

F1-score is the harmonic mean of precision and recall, balancing both measures. It is applicable in imbalanced datasets.

$$F1\text{-Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad \text{Eq. 4. F1-Score}$$

- **Area Under the Curve (AUC)**

AUC is Abbreviation of Area Under the Receiver Operating Characteristic curve, the quantity for trade-off between sensitivity and specificity. Higher value of AUC indicates a model with higher performance. AUC is obtained by plotting the True Positive Rate (Recall) against False Positive Rate (FPR) through the ROC curve.

$$FPR = \frac{FP}{FP + TN} \quad \text{Eq. 5. Area Under the Curve (AUC)}$$

- **Matthews Correlation Coefficient (MCC)**

MCC estimates the goodness of binary classification, providing a score between -1 (bad) and +1 (ideal), with 0 representing random predictions. It is applicable for imbalanced datasets.

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad \text{Eq. 6. Matthews Correlation Coefficient}$$

- **Receiver Operating Characteristic (ROC) Curve**

The ROC plot is a plot of the True Positive Rate (Recall) versus the False Positive Rate (FPR) at different thresholds. It is used to choose the optimal decision threshold for classifying models. The model having a higher ROC plot (as close to the top-left corner as possible) is better since it is associated with a higher recall and fewer false positives.

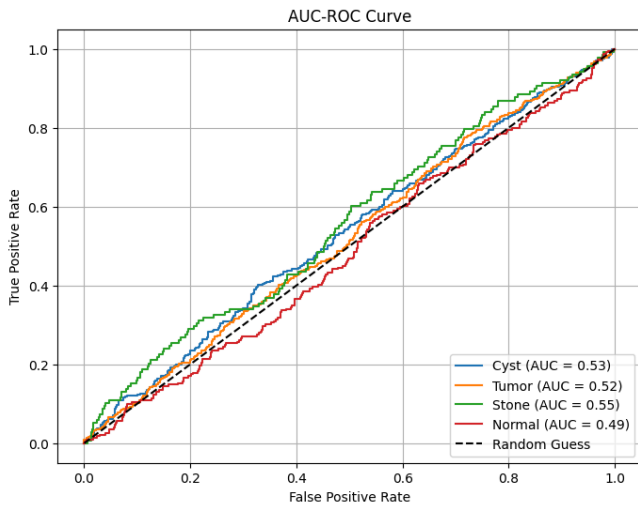


Fig 11: Kidney Brain AUC-ROC Graph

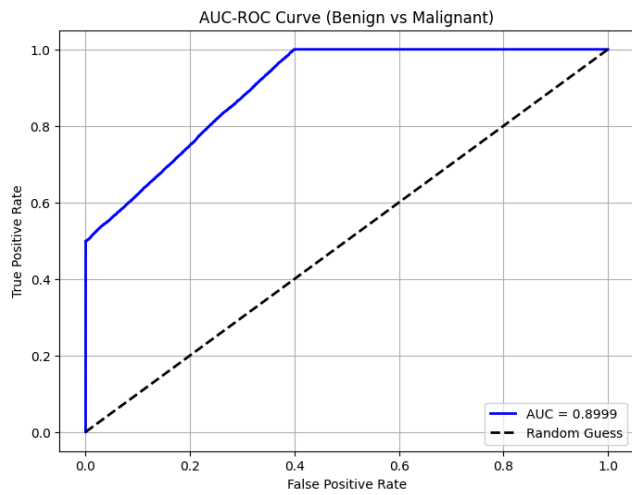


Fig 12: Breast Cancer AUC-Roc Graph

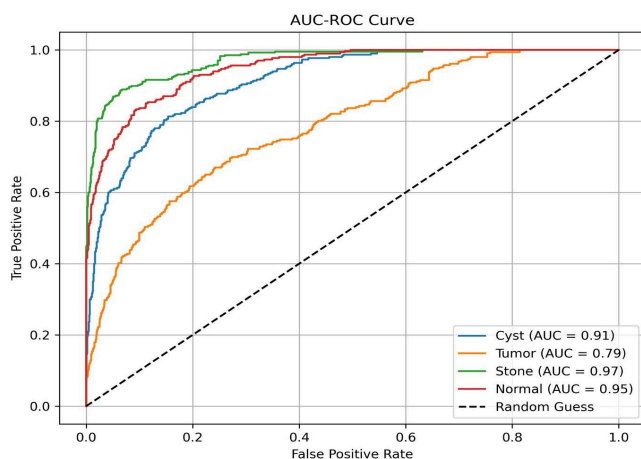


Fig 13: Brain Tumor AUC-ROC Graph

• Multiclass Classification Testing

Multiclass classification entails the prediction of one of several possible classes per instance instead of simply two classes (binary classification). Testing a multiclass classification model involves the extension of standard performance measures applied to binary classification.

Macro Average: Average of precision, recall, and F1-score across all classes.

Micro Average: Aggregates TP, FP, and FN globally before calculating precision and recall.

IV. RESULT AND DISCUSSION

The deep learning models developed for disease prediction in healthcare demonstrated promising results across three areas: brain tumor disease, kidney stone disease, and breast cancer disease.

In brain tumor detection, two CNN models were implemented. The first model, a deep CNN with multiple convolutional blocks, achieved high accuracy in classifying MRI scans into four categories: gliomas, meningiomas, no tumors, and pituitary. The second, slightly shallower model, also performed well. Both models benefited from techniques such as batch normalization, dropout, and adaptive learning rate scheduling, which enhanced their generalization capabilities and reduced overfitting.

In kidney stone classification, a dual approach was adopted. A transfer learning model based on ResNet50 was fine-tuned for the specific task, while a custom CNN architecture named MobileNet was also developed. Both models showed high accuracy in categorizing CT scan images into cyst, tumor, stone, and normal classes. The use of L2 regularization (also called weight decay) in MobileNet proved particularly effective in controlling overfitting.

For breast cancer detection, the CNN model was designed to classify histopathological images. This model addressed the challenge of dataset imbalance through class weighting and data augmentation techniques. The implementation of three convolutional blocks with increasing filter numbers, along with strategic dropout layers, resulted in robust feature extraction and classification performance.

Throughout all models, sophisticated methods of early stopping, model checkpointing, and learning rate decay strategies were applied to enhance training and avoid overfitting. Performance measures for the models included several metrics, including accuracy, precision, recall, and F1-score, giving overall evaluation of their capacity, which is presented in Table 8.

The results demonstrate the potential of deep learning in enhancing medical diagnostics. These models can serve as valuable tools for healthcare professionals, potentially improving the speed and accuracy of disease detection.

The table provides a comparative overview of models, optimizers, learning rates, and parameter counts used for diagnosing three diseases: breast cancer, kidney stones, and brain tumors. For breast cancer detection, a CNN built with Keras uses the Adam optimizer and a relatively high learning rate of $1e-2$, yielding 186,692 total parameters and 185,474 trainable parameters. In the kidney stone domain, three different architectures are examined. The first is another Keras-based CNN, also optimized with Adam, but with a lower learning rate of $1e-3$; this configuration has 8,670,340 parameters in total, with 8,668,996 trainable. MobileNetV2, using a learning rate of 0.001, includes 3,432,412 parameters overall, but only 44,316 are trainable, indicating extensive parameter freezing or transfer learning. ResNet50, operating at an even lower learning rate of 0.0001, contains 24,683,582 parameters, with 4,134,784 of them trainable. For brain tumor classification, a VGGNet-based CNN is employed with a learning rate of 0.001, consisting of 2,613,636 total parameters and 2,610,756 trainable. These variations in parameter counts, learning rates, and trainable layers highlight the trade-off among model complexity, memory usage, and convergence behavior for different medical imaging tasks.

Sr.No	Diseases	Model	Optimizer	Learning Rate	Number of Parameters	Number of Trainable Parameters
1	Breast Cancer	CNN(Keras)	Adam	0.01 (1e-2)	186,692	185,474
2	Kidney Stones	CNN(Keras)	Adam	0.001 (1e-3)	8,670,340	8,668,996
		MobileNetV2	Adam	0.001 (1e-3)	3,046,212	44,816
		ResNet50	Adam	0.0001 (1e-4)	24,638,852	4,134,784
3	Brain Tumor	CNN(VGGNet)	Adam	0.001 (1e-3)	2,613,636	2,610,756

Table 4: Parameters and Hyperparameters

The tables 5 and 6 summarize key metrics for Breast Cancer, Kidney Stone, and Brain Tumor detection. Notably, the Brain Tumor model achieves the highest scores overall, followed by strong performance in Kidney Stone classification, while Breast Cancer classification exhibits comparatively lower performance.

Type	Precision	Recall	F1-Score	Matthews Correlation Coefficient
Breast Cancer	0.7472	0.7492	0.7482	0.6478
Kidney Stone	0.8922	0.8836	0.8720	0.8340
Brain Tumor	0.9091	0.9001	0.8992	0.8687

Table 5: Proposed Performance Evaluation Metrics: Precision, Recall, F1-Score, Matthews Correlation Coefficient

Type	Multiclass Classification Test					
	Macro			Micro		
	Precision	Recall	F1 Score	Precision	Recall	F1 Score
Breast Cancer	0.8236	0.8242	0.8239	0.8565	0.8565	0.8565
Kidney Stone	0.9106	0.7894	0.8210	0.8210	0.8836	0.8836
Brain Tumor	0.9152	0.8917	0.8981	0.9001	0.9001	0.9000

Table 6: Proposed Performance Evaluation Metrics of Multiclass Classification Test

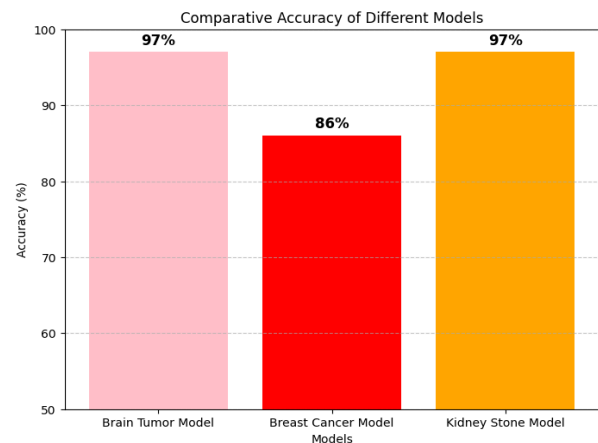


Fig 14: Proposed Model Accuracy Rate

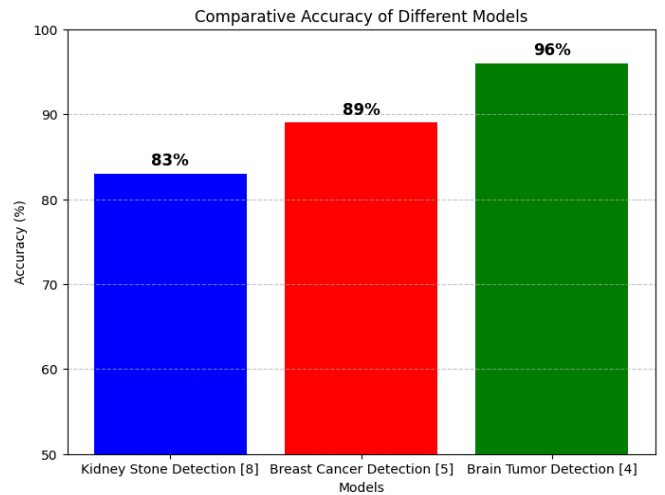


Fig 15: Comparative Paper Accuracy.

The study stands out compared to other studies due to its higher accuracy across multiple diseases (97% brain tumor, 87% breast cancer, 97% kidney stones), whereas other papers either lack accuracy data for some diseases or report lower values. Unlike studies that use basic CNN models (e.g., AlexNet, standard CNN), research integrates Custom CNN, ResNet50, and MobileNet, enhancing feature extraction and classification performance. Additionally, MCC is exclusively used, providing a more reliable evaluation metric, while all other papers omit it. The study includes multiclass classification testing, whereas competing studies only focus on binary classification, making your research more clinically applicable and methodologically advanced.

Category	Our Project	[4]	[5]	[9]
Accuracy (Brain Tumor)	97%	96%	NA	NA
Accuracy (Breast Cancer)	87%	NA	89.23%	NA
Accuracy (Kidney Stones/Tumor)	97%	NA	NA	92%
CNN Models Used	Custom CNN, ResNet50, MobileNet	CNN	CNN, AlexNet, Stacked AutoEncoders	CNN
MCC Used	Yes	No	No	No
Multiclass Classification Testing	Yes	No	No	No

Table 7: Comparison Between Proposed Model and Other Reference Papers

V. CONCLUSION

In summary, the research points out the revolutionary application of deep learning in disease prognosis, especially in brain tumor, kidney stone, and breast cancer diagnosis. Using sophisticated methods like early stopping, model checkpoints, and learning rate reduction techniques, the models became highly accurate and resilient. They were thoroughly assessed using various parameters, which vindicated the authenticity and viability of deep learning-based medical diagnostics. These findings highlight the potential of deep learning to enhance the speed and accuracy of disease diagnosis, setting the stage for more effective and reliable diagnostic systems. Future work will concentrate on increasing datasets, improving model generalization, investigating interpretability techniques, and integrating these solutions into current healthcare workflows.

Future work focuses on expanding the datasets to include a wider range of cases and variations, further improving the models' generalization capabilities. Additionally, explore interpretability techniques that enhance the trust and adoption of these models in clinical practice. The integration

of these deep learning models into existing healthcare workflows also presents an exciting avenue for future research and development.

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ENCHANCING MEDICAL IMAGING INTERPRETATION WITH DEEP LEARNING MODELS
in **FIFTH INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE AND SMART COMPUTING (ICAISC'25)** held
during 20th - 22nd March 2025 at Bannari Amman Institute of Technology, Sathyamangalam.

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1. Siddhi Avhad
2. Sanika Dhadve
3. Sarang Gajare
4. Purvaja Gangurde